

The Notebooks · Era VI — Change

Dark green lab notebook

Testing Mode · UK · IB · US

Exercises

T.VI.1.001 · AP FRQ: limit justification from table/graph · challenge · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «AP FRQ: limit justification from table/graph»?
- (b) [allowed/US/2mk] Write a justification sentence frame for «AP FRQ: limit justification from table/graph»: Because ____, by ____, therefore ____.
- (c) [allowed/US/3mk] Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «AP FRQ: limit justification from table/graph», then interpret.

T.VI.1.002 · Formal limit language where AP demands precision · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/2mk] Evaluate $\lim_{x \rightarrow 6} (x^2 - 36)/(x - 6)$.
- (b) [allowed/US/3mk] IVT: f continuous on $[6, 9]$, $f(6) = -2$, $f(9) = 5$. Justify a root of $f(x) = 0$.
- (c) [allowed/US/2mk] Distinguish a vertical asymptote from a removable hole in one sentence.

T.VI.1.003 · IVT justification FRQ stems · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «IVT justification FRQ stems»?
- (b) [allowed/US/2mk] Write a justification sentence frame for «IVT justification FRQ stems»: Because ____, by ____, therefore ____.
- (c) [allowed/US/3mk] Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «IVT justification FRQ stems», then interpret.

T.VI.1.004 · Continuity checklist FRQ · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «Continuity checklist FRQ»?
- (b) [allowed/US/2mk] Write a justification sentence frame for «Continuity checklist FRQ»: Because ____, by ____, therefore ____.
- (c) [allowed/US/3mk] Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Continuity checklist FRQ», then interpret.

T.VI.1.005 · Limits analytically vs numerically GDC · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/2mk] Evaluate $\lim_{x \rightarrow 3} (x^2 - 9)/(x - 3)$.
- (b) [gdc/US/3mk] IVT: f continuous on $[3, 6]$, $f(3) = -2$, $f(6) = 5$. Justify a root of $f(x) = 0$.
- (c) [gdc/US/3mk ■ LAB] GDC lab for «Limits analytically vs numerically GDC»: record menu path and numerical output.

T.VI.1.006 · Squeeze theorem justification write-ups · challenge · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «Squeeze theorem justification write-ups»?
- (b) [allowed/US/2mk] Write a justification sentence frame for «Squeeze theorem justification write-ups»: Because ____, by ____, therefore ____.
- (c) [allowed/US/3mk] Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Squeeze theorem justification write-ups», then interpret.

T.VI.1.007 · Horizontal asymptotes via limits at infinity FRQ · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «Horizontal asymptotes via limits at infinity FRQ»?
- (b) [allowed/US/2mk] Write a justification sentence frame for «Horizontal asymptotes via limits at infinity FRQ»: Because ____, by ____, therefore ____.
- (c) [allowed/US/3mk ■ LAB] Desmos lab for «Horizontal asymptotes via limits at infinity FRQ»: vary one parameter, capture before/after, write one observation.

T.VI.1.008 · Vertical asymptotes vs holes FRQ language · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «Vertical asymptotes vs holes FRQ language»?
- (b) [allowed/US/2mk] Write a justification sentence frame for «Vertical asymptotes vs holes FRQ language»: Because ____, by ____, therefore ____.
- (c) [allowed/US/3mk] Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Vertical asymptotes vs holes FRQ language», then interpret.

T.VI.1.009 · Epsilon-N / epsilon-delta light AP+ · strange · AP_CALC_BC,IB_AA_HL,IB_AA_HL

- (a) [never/neutral/2mk] Evaluate $\lim_{x \rightarrow 4} (x^2 - 16)/(x - 4)$.
- (b) [allowed/US/3mk] IVT: f continuous on $[4, 7]$, $f(4) = -2$, $f(7) = 5$. Justify a root of $f(x) = 0$.
- (c) [allowed/US/2mk] Distinguish a vertical asymptote from a removable hole in one sentence.

T.VI.1.010 · One-sided limits for piecewise continuity · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/2mk] Evaluate $\lim_{x \rightarrow 6} (x^2 - 36)/(x - 6)$.
- (b) [allowed/US/3mk] IVT: f continuous on $[6, 9]$, $f(6) = -2$, $f(9) = 5$. Justify a root of $f(x) = 0$.
- (c) [allowed/US/2mk] Distinguish a vertical asymptote from a removable hole in one sentence.

T.VI.2.001 · AP FRQ: definition of derivative as limit · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «AP FRQ: definition of derivative as limit»?
- (b) [allowed/US/2mk] Write a justification sentence frame for «AP FRQ: definition of derivative as limit»: Because ____, by ____, therefore ____.
- (c) [allowed/US/3mk] Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «AP FRQ: definition of derivative as limit», then interpret.

T.VI.2.002 · Differentiability vs continuity FRQ traps · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «Differentiability vs continuity FRQ traps»?
- (b) [allowed/US/2mk] Write a justification sentence frame for «Differentiability vs continuity FRQ traps»: Because ____, by ____, therefore ____.
- (c) [allowed/US/3mk] Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Differentiability vs continuity FRQ traps», then interpret.

T.VI.2.003 · L'Hôpital packaging as named AP topic · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/1mk] Differentiate $y = 4x^4$.
- (b) [never/US/2mk] $f(x) = 4x^4$. Find $f'(2)$.
- (c) [never/US/2mk] $s(t) = 4t^2$ metres. Velocity at $t = 4$ s?

T.VI.2.004 · Implicit differentiation AP FRQ · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «Implicit differentiation AP FRQ»?
- (b) [allowed/US/2mk] Write a justification sentence frame for «Implicit differentiation AP FRQ»: Because ____, by ____, therefore ____.
- (c) [allowed/US/3mk] Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Implicit differentiation AP FRQ», then interpret.

T.VI.2.005 · Related rates setup sentence frames · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/1mk] Differentiate $y = 5x^2$.
- (b) [never/US/2mk] $f(x) = 5x^2$. Find $f'(2)$.
- (c) [never/US/2mk] $s(t) = 5t^2$ metres. Velocity at $t = 2$ s?

T.VI.2.006 · Calculator-active derivative estimates · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/1mk] Differentiate $y = 6x^2$.
- (b) [never/US/2mk] $f(x) = 6x^2$. Find $f'(2)$.
- (c) [gdc/US/3mk ■ LAB] GDC lab for «Calculator-active derivative estimates»: record menu path and numerical output.

T.VI.2.007 · Chain rule layered AP items · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «Chain rule layered AP items»?
- (b) [allowed/US/2mk] Write a justification sentence frame for «Chain rule layered AP items»: Because ____, by ____, therefore ____.
- (c) [allowed/US/3mk] Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Chain rule layered AP items», then interpret.

T.VI.2.008 · Second derivative from parametric (BC) · challenge · AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/1mk] Differentiate $y = 7x^4$.
- (b) [never/US/2mk] $f(x) = 7x^4$. Find $f'(2)$.
- (c) [never/US/2mk] $s(t) = 7t^2$ metres. Velocity at $t = 4$ s?

T.VI.2.009 · Logarithmic differentiation AP items · challenge · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/1mk] Differentiate $y = 7x^3$.
- (b) [never/US/2mk] $f(x) = 7x^3$. Find $f'(2)$.
- (c) [never/US/2mk] $s(t) = 7t^2$ metres. Velocity at $t = 3$ s?

T.VI.2.010 · Motion along a line: when particle at rest · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/1mk] Differentiate $y=7x^2$.
- (b) [never/US/2mk] $f(x)=7x^2$. Find $f'(2)$.
- (c) [never/US/2mk] $s(t)=7t^2$ metres. Velocity at $t=2$ s?

T.VI.2.011 · Approximate derivative from table FRQ · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «Approximate derivative from table FRQ»?
- (b) [allowed/US/2mk] Write a justification sentence frame for «Approximate derivative from table FRQ»: Because ____, by ____, therefore ____.
- (c) [allowed/US/3mk] Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Approximate derivative from table FRQ», then interpret.

T.VI.3.001 · MVT justification FRQ stems · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «MVT justification FRQ stems»?
- (b) [allowed/US/2mk] Write a justification sentence frame for «MVT justification FRQ stems»: Because ____, by ____, therefore ____.
- (c) [allowed/US/3mk] Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «MVT justification FRQ stems», then interpret.

T.VI.3.002 · EVT / Extreme Value Theorem justification · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «EVT / Extreme Value Theorem justification»?
- (b) [allowed/US/2mk] Write a justification sentence frame for «EVT / Extreme Value Theorem justification»: Because ____, by ____, therefore ____.
- (c) [allowed/US/3mk] Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «EVT / Extreme Value Theorem justification», then interpret.

T.VI.3.003 · Candidates test write-up discipline · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/1mk] Differentiate $y=4x^2$.
- (b) [never/US/2mk] $f(x)=4x^2$. Find $f'(2)$.
- (c) [never/US/2mk] $s(t)=4t^2$ metres. Velocity at $t=2$ s?

T.VI.3.004 · f, f', f'' sign chart FRQ classic · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «f, f', f'' sign chart FRQ classic»?
- (b) [allowed/US/2mk] Write a justification sentence frame for «f, f', f'' sign chart FRQ classic»: Because ____, by ____, therefore ____.
- (c) [allowed/US/3mk] ■ LAB] Desmos lab for «f, f', f'' sign chart FRQ classic»: vary one parameter, capture before/after, write one observation.

T.VI.3.005 · Optimization AP contextual FRQ · challenge · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «Optimization AP contextual FRQ»?
- (b) [allowed/US/2mk] Write a justification sentence frame for «Optimization AP contextual FRQ»: Because ____, by ____, therefore ____.
- (c) [allowed/US/3mk] Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Optimization AP contextual FRQ», then interpret.

T.VI.3.006 · Linearisation / tangent approx error talk · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/1mk] Differentiate $y=5x^4$.
- (b) [never/US/2mk] $f(x)=5x^4$. Find $f'(2)$.
- (c) [never/US/2mk] $s(t)=5t^2$ metres. Velocity at $t=4$ s?

T.VI.3.007 · Particle motion: velocity/acceleration FRQ · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «Particle motion: velocity/acceleration FRQ»?
- (b) [allowed/US/2mk] Write a justification sentence frame for «Particle motion: velocity/acceleration FRQ»: Because ____, by ____, therefore ____.
- (c) [allowed/US/3mk] Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Particle motion: velocity/acceleration FRQ», then interpret.

T.VI.3.008 · Newton's method AP/IB numerical · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/1mk] Differentiate $y=6x^5$.
- (b) [never/US/2mk] $f(x)=6x^5$. Find $f'(2)$.
- (c) [gdc/US/3mk] ■ LAB] GDC lab for «Newton's method AP/IB numerical»: record menu path and numerical output.

T.VI.3.009 · Related rates classic cone/ladder/shadow · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/1mk] Differentiate $y=2x^2$.
- (b) [never/US/2mk] $f(x)=2x^2$. Find $f'(2)$.
- (c) [never/US/2mk] $s(t)=2t^2$ metres. Velocity at $t=2$ s?

T.VI.3.010 · Concavity and inflection FRQ write-ups · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «Concavity and inflection FRQ write-ups»?
- (b) [allowed/US/2mk] Write a justification sentence frame for «Concavity and inflection FRQ write-ups»: Because ____, by ____, therefore ____.
- (c) [allowed/US/3mk ■ LAB] Desmos lab for «Concavity and inflection FRQ write-ups»: vary one parameter, capture before/after, write one observation.

T.VI.3.011 · L'Hôpital repeated applications · challenge · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/1mk] Differentiate $y=3x^4$.
- (b) [never/US/2mk] $f(x)=3x^4$. Find $f'(2)$.
- (c) [never/US/2mk] $s(t)=3t^2$ metres. Velocity at $t=4$ s?

T.VI.3.012 · Graphing calculator: find zeros of f · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/1mk] Differentiate $y=4x^3$.
- (b) [never/US/2mk] $f(x)=4x^3$. Find $f'(2)$.
- (c) [gdc/US/3mk ■ LAB] GDC lab for «Graphing calculator: find zeros of f»: record menu path and numerical output.

T.VI.3.013 · AP calculator-active vs inactive practice modes · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/1mk] Differentiate $y=5x^3$.
- (b) [never/US/2mk] $f(x)=5x^3$. Find $f'(2)$.
- (c) [gdc/US/3mk ■ LAB] GDC lab for «AP calculator-active vs inactive practice modes»: record menu path and numerical output.

T.VI.3.014 · Absolute extrema on closed interval FRQ · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «Absolute extrema on closed interval FRQ»?
- (b) [allowed/US/2mk] Write a justification sentence frame for «Absolute extrema on closed interval FRQ»: Because ____, by ____, therefore ____.
- (c) [allowed/US/3mk] Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Absolute extrema on closed interval FRQ», then interpret.

T.VI.4.001 · Riemann sum notation left/right/mid FRQ · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «Riemann sum notation left/right/mid FRQ»?
- (b) [allowed/US/2mk] Write a justification sentence frame for «Riemann sum notation left/right/mid FRQ»: Because ____, by ____, therefore ____.
- (c) [allowed/US/3mk] Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Riemann sum notation left/right/mid FRQ», then interpret.

T.VI.4.002 · FTC graphical accumulation FRQ · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «FTC graphical accumulation FRQ»?
- (b) [allowed/US/2mk] Write a justification sentence frame for «FTC graphical accumulation FRQ»: Because ____, by ____, therefore ____.
- (c) [allowed/US/3mk ■ LAB] Graph lab for «FTC graphical accumulation FRQ»: plot, mark key points, state the trend.

T.VI.4.003 · Average value theorem applications · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/1mk] $\int 5x^4 dx$.
- (b) [never/US/2mk] Evaluate $\int_{-1}^1 5x dx$.
- (c) [allowed/US/2mk] If $v(t)$ is velocity, what does $\int_{-1}^1 v(t) dt$ represent?

T.VI.4.004 · Definite integral properties FRQ · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «Definite integral properties FRQ»?
- (b) [allowed/US/2mk] Write a justification sentence frame for «Definite integral properties FRQ»: Because ____, by ____, therefore ____.
- (c) [allowed/US/3mk] Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Definite integral properties FRQ», then interpret.

T.VI.4.005 · Trapezoidal rule approximation FRQ · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «Trapezoidal rule approximation FRQ»?
- (b) [allowed/US/2mk] Write a justification sentence frame for «Trapezoidal rule approximation FRQ»: Because ____, by ____, therefore ____.
- (c) [allowed/US/3mk] Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Trapezoidal rule approximation FRQ», then interpret.

T.VI.4.006 · FTC derivative of integral with variable limits · challenge · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/1mk] $\int 6x^4 dx$.
- (b) [never/US/2mk] Evaluate $\int_{-1}^1 6x dx$.
- (c) [allowed/US/2mk] If $v(t)$ is velocity, what does $\int_{-1}^1 v(t) dt$ represent?

T.VI.4.007 · AP numerical integration on GDC · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/1mk] $\int 5x^3 dx$.
- (b) [never/US/2mk] Evaluate $\int_{-1}^1 5x dx$.
- (c) [gdc/US/3mk ■ LAB] GDC lab for «AP numerical integration on GDC»: record menu path and numerical output.

T.VI.4.008 · Accumulation from rate-in rate-out FRQ · challenge · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «Accumulation from rate-in rate-out FRQ»?
- (b) [allowed/US/2mk] Write a justification sentence frame for «Accumulation from rate-in rate-out FRQ»: Because ____, by ____, therefore ____.
- (c) [allowed/US/3mk ■ LAB] Graph lab for «Accumulation from rate-in rate-out FRQ»: plot, mark key points, state the trend.

T.VI.4.009 · Signed area vs total area language · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/1mk] $\int 6x^2 dx$.
- (b) [never/US/2mk] Evaluate $\int_{-1}^1 6x dx$.
- (c) [allowed/US/3mk ■ LAB] Graph lab for «Signed area vs total area language»: plot, mark key points, state the trend.

T.VI.5.001 · Integration by parts IB AA / AP BC · core · AP_CALC_BC,IB_AA_HL,IB_AA_HL

- (a) [never/neutral/1mk] $\int 5x^3 dx$.
- (b) [never/US/2mk] Evaluate $\int_{-1}^1 5x dx$.
- (c) [allowed/US/2mk] If $v(t)$ is velocity, what does $\int_{-1}^1 v(t) dt$ represent?

T.VI.5.002 · Partial fractions integration exam · challenge · AP_CALC_BC,IB_AA_HL,IB_AA_HL

- (a) [never/neutral/1mk] $\int 2x^4 dx$.
- (b) [never/US/2mk] Evaluate $\int_{-1}^1 2x dx$.
- (c) [allowed/US/2mk] If $v(t)$ is velocity, what does $\int_{-1}^1 v(t) dt$ represent?

T.VI.5.003 · Improper integrals convergence tests light · challenge · AP_CALC_BC,IB_AA_HL,IB_AA_HL

- (a) [never/neutral/1mk] $\int 4x^4 dx$.
- (b) [never/US/2mk] Evaluate $\int_{-1}^1 4x dx$.
- (c) [allowed/US/2mk] If $v(t)$ is velocity, what does $\int_{-1}^1 v(t) dt$ represent?

T.VI.5.004 · Trigonometric integrals powers of sin/cos · challenge · AP_CALC_BC,IB_AA_HL,IB_AA_HL

- (a) [never/neutral/1mk] $\int 2x^1 dx$.
- (b) [never/US/2mk] Evaluate $\int_{-1}^1 2x dx$.
- (c) [allowed/US/2mk] If $v(t)$ is velocity, what does $\int_{-1}^1 v(t) dt$ represent?

T.VI.5.005 · Trigonometric substitution light (BC/HL) · strange · AP_CALC_BC,IB_AA_HL,IB_AA_HL

- (a) [never/neutral/1mk] $\int 4x^4 dx$.
- (b) [never/US/2mk] Evaluate $\int_{-1}^1 4x dx$.
- (c) [allowed/US/2mk] If $v(t)$ is velocity, what does $\int_{-1}^1 v(t) dt$ represent?

T.VI.6.001 · Area between curves AP FRQ classic · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/2mk] 150 cm to nearest 10 cm. Write the error interval for length L.
- (b) [never/US/2mk] $w=6.0$ cm, $l=9.5$ cm each to nearest 0.1 cm. Upper bound of perimeter?
- (c) [never/US/2mk] Maximise speed = distance/time given rounded measurements. Which bounds?

T.VI.6.002 · Disk/washer volume FRQ · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «Disk/washer volume FRQ»?
- (b) [allowed/US/2mk] Write a justification sentence frame for «Disk/washer volume FRQ»: Because ___, by ___, therefore ___.
- (c) [allowed/US/3mk] Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Disk/washer volume FRQ», then interpret.

T.VI.6.003 · Known cross-section volume FRQ · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «Known cross-section volume FRQ»?
- (b) [allowed/US/2mk] Write a justification sentence frame for «Known cross-section volume FRQ»: Because ___, by ___, therefore ___.
- (c) [allowed/US/3mk] Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Known cross-section volume FRQ», then interpret.

T.VI.6.004 · Net change theorem worded FRQ · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «Net change theorem worded FRQ»?
- (b) [allowed/US/2mk] Write a justification sentence frame for «Net change theorem worded FRQ»: Because ___, by ___, therefore ___.
- (c) [allowed/US/3mk] Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Net change theorem worded FRQ», then interpret.

T.VI.6.005 · Arc length setup (BC) · core · AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/1mk] Differentiate $y=5x^5$.
- (b) [never/US/2mk] $f(x)=5x^5$. Find $f'(2)$.
- (c) [never/US/2mk] $s(t)=5t^2$ metres. Velocity at $t=5$ s?

T.VI.6.006 · Shell method when washer awkward · challenge · AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/1mk] Differentiate $y=6x^4$.
- (b) [never/US/2mk] $f(x)=6x^4$. Find $f'(2)$.
- (c) [never/US/2mk] $s(t)=6t^2$ metres. Velocity at $t=4$ s?

T.VI.6.007 · Work pumping liquid FRQ style · challenge · AP_CALC_BC,IB_AA_HL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «Work pumping liquid FRQ style»?
- (b) [allowed/US/2mk] Write a justification sentence frame for «Work pumping liquid FRQ style»: Because ___, by ___, therefore ___.
- (c) [allowed/US/3mk] Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Work pumping liquid FRQ style», then interpret.

T.VI.6.008 · Probability density integral applications · strange · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/1mk] Differentiate $y=5x^5$.
- (b) [never/US/2mk] $f(x)=5x^5$. Find $f'(2)$.
- (c) [never/US/2mk] $s(t)=5t^2$ metres. Velocity at $t=5$ s?

T.VI.6.009 · Displacement from velocity graph areas · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/1mk] Differentiate $y=2x^4$.
- (b) [never/US/2mk] $f(x)=2x^4$. Find $f'(2)$.
- (c) [allowed/US/3mk ■ LAB] Graph lab for «Displacement from velocity graph areas»: plot, mark key points, state the trend.

T.VI.7.001 · Slope fields sketch and match FRQ · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «Slope fields sketch and match FRQ»?
- (b) [allowed/US/2mk] Write a justification sentence frame for «Slope fields sketch and match FRQ»: Because ___, by ___, therefore ___.
- (c) [allowed/US/3mk ■ LAB] Desmos lab for «Slope fields sketch and match FRQ»: vary one parameter, capture before/after, write one observation.

T.VI.7.002 · Euler's method table FRQ · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «Euler's method table FRQ»?
- (b) [allowed/US/2mk] Write a justification sentence frame for «Euler's method table FRQ»: Because ___, by ___, therefore ___.
- (c) [allowed/US/3mk] Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Euler's method table FRQ», then interpret.

T.VI.7.003 · Logistic DE: $\frac{dP}{dt}=kP(M-P)$ exam pack · challenge · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/1mk] Differentiate $y=6x^4$.
- (b) [never/US/2mk] $f(x)=6x^4$. Find $f'(2)$.
- (c) [never/US/2mk] $s(t)=6t^2$ metres. Velocity at $t=4$ s?

T.VI.7.004 · Separable DE with initial condition FRQ · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «Separable DE with initial condition FRQ»?
- (b) [allowed/US/2mk] Write a justification sentence frame for «Separable DE with initial condition FRQ»: Because ___, by ___, therefore ___.
- (c) [allowed/US/3mk] Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Separable DE with initial condition FRQ», then interpret.

T.VI.7.005 · Growth/decay DE contextual FRQ · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «Growth/decay DE contextual FRQ»?
- (b) [allowed/US/2mk] Write a justification sentence frame for «Growth/decay DE contextual FRQ»: Because ___, by ___, therefore ___.
- (c) [allowed/US/3mk] Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Growth/decay DE contextual FRQ», then interpret.

T.VI.7.006 · Newton cooling DE exam · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/1mk] Differentiate $y=7x^3$.
- (b) [never/US/2mk] $f(x)=7x^3$. Find $f'(2)$.
- (c) [never/US/2mk] $s(t)=7t^2$ metres. Velocity at $t=3$ s?

T.VI.7.007 · Mixing problems DE setup · challenge · AP_CALC_BC,IB_AA_HL,IB_AA_HL

- (a) [never/neutral/1mk] Differentiate $y=7x^3$.
- (b) [never/US/2mk] $f(x)=7x^3$. Find $f'(2)$.
- (c) [never/US/2mk] $s(t)=7t^2$ metres. Velocity at $t=3$ s?

T.VI.7.008 · Equilibrium solutions stability talk · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/1mk] Differentiate $y=4x^5$.
- (b) [never/US/2mk] $f(x)=4x^5$. Find $f'(2)$.
- (c) [never/US/2mk] $s(t)=4t^2$ metres. Velocity at $t=5$ s?

T.VI.7.009 · IB DE modelling emphasis pack · challenge · IB_AI_SL,IB_AI_HL,IB_AA_SL

- (a) [never/neutral/1mk] Differentiate $y=7x^4$.
- (b) [never/IB/2mk] $f(x)=7x^4$. Find $f'(2)$.
- (c) [never/IB/2mk] $s(t)=7t^2$ metres. Velocity at $t=4$ s?

T.VI.7.010 · Particular vs general solution language · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/1mk] Differentiate $y=2x^4$.
- (b) [never/US/2mk] $f(x)=2x^4$. Find $f'(2)$.
- (c) [never/US/2mk] $s(t)=2t^2$ metres. Velocity at $t=4$ s?

T.VI.7.011 · Euler vs analytic solution comparison · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/1mk] Differentiate $y=5x^5$.
- (b) [never/US/2mk] $f(x)=5x^5$. Find $f'(2)$.
- (c) [gdc/US/3mk ■ LAB] GDC lab for «Euler vs analytic solution comparison»: record menu path and numerical output.

T.VI.7.012 · Logistic carrying capacity interpretation · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/1mk] Differentiate $y=3x^4$.
- (b) [never/US/2mk] $f(x)=3x^4$. Find $f'(2)$.
- (c) [allowed/US/3mk ■ LAB] Desmos lab for «Logistic carrying capacity interpretation»: vary one parameter, capture before/after, write one observation.

T.VI.7.013 · Slope field: draw solution through a point · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/1mk] Differentiate $y=4x^2$.
- (b) [never/US/2mk] $f(x)=4x^2$. Find $f'(2)$.
- (c) [allowed/US/3mk ■ LAB] Desmos lab for «Slope field: draw solution through a point»: vary one parameter, capture before/after, write one observation.

T.VI.7.014 · Euler method stepped table 1 · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/1mk] Differentiate $y=5x^2$.
- (b) [never/US/2mk] $f(x)=5x^2$. Find $f'(2)$.
- (c) [never/US/2mk] $s(t)=5t^2$ metres. Velocity at $t=2$ s?

T.VI.7.015 · Logistic model interpretation 1 · challenge · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/1mk] Differentiate $y=3x^3$.
- (b) [never/US/2mk] $f(x)=3x^3$. Find $f'(2)$.
- (c) [allowed/US/3mk ■ LAB] Desmos lab for «Logistic model interpretation 1»: vary one parameter, capture before/after, write one observation.

T.VI.7.016 · Euler method stepped table 2 · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/1mk] Differentiate $y=4x^2$.
- (b) [never/US/2mk] $f(x)=4x^2$. Find $f'(2)$.
- (c) [never/US/2mk] $s(t)=4t^2$ metres. Velocity at $t=2$ s?

T.VI.7.017 · Logistic model interpretation 2 · challenge · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/1mk] Differentiate $y=2x^3$.
- (b) [never/US/2mk] $f(x)=2x^3$. Find $f'(2)$.
- (c) [allowed/US/3mk ■ LAB] Desmos lab for «Logistic model interpretation 2»: vary one parameter, capture before/after, write one observation.

T.VI.7.018 · Euler method stepped table 3 · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/1mk] Differentiate $y=7x^3$.
- (b) [never/US/2mk] $f(x)=7x^3$. Find $f'(2)$.
- (c) [never/US/2mk] $s(t)=7t^2$ metres. Velocity at $t=3$ s?

T.VI.7.019 · Logistic model interpretation 3 · challenge · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/1mk] Differentiate $y=6x^4$.
- (b) [never/US/2mk] $f(x)=6x^4$. Find $f'(2)$.
- (c) [allowed/US/3mk ■ LAB] Desmos lab for «Logistic model interpretation 3»: vary one parameter, capture before/after, write one observation.

T.VI.8.001 · Parametric derivatives dy/dx FRQ · core · AP_CALC_BC,IB_AA_HL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «Parametric derivatives dy/dx FRQ»?
- (b) [allowed/US/2mk] Write a justification sentence frame for «Parametric derivatives dy/dx FRQ»: Because ___, by ___, therefore ___.
- (c) [allowed/US/3mk] Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Parametric derivatives dy/dx FRQ», then interpret.

T.VI.8.002 · Parametric arc length FRQ · core · AP_CALC_BC,IB_AA_HL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «Parametric arc length FRQ»?
- (b) [allowed/US/2mk] Write a justification sentence frame for «Parametric arc length FRQ»: Because ___, by ___, therefore ___.
- (c) [allowed/US/3mk] Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Parametric arc length FRQ», then interpret.

T.VI.8.003 · Particle path parametric motion FRQ · core · AP_CALC_BC,IB_AA_HL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «Particle path parametric motion FRQ»?
- (b) [allowed/US/2mk] Write a justification sentence frame for «Particle path parametric motion FRQ»: Because ___, by ___, therefore ___.
- (c) [allowed/US/3mk] Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Particle path parametric motion FRQ», then interpret.

T.VI.8.004 · Polar area $\frac{1}{2}r^2 d\theta$ FRQ · core · AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/2mk] 170 cm to nearest 10 cm. Write the error interval for length L.
- (b) [never/US/2mk] $w=4.8$ cm, $l=11.2$ cm each to nearest 0.1 cm. Upper bound of perimeter?
- (c) [never/US/2mk] Maximise speed = distance/time given rounded measurements. Which bounds?

T.VI.8.005 · Polar slope dy/dx in polar form · challenge · AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/1mk] $\int 2x^3 dx$.
- (b) [never/US/2mk] Evaluate $\int_{-1}^1 2x dx$.
- (c) [allowed/US/2mk] If $v(t)$ is velocity, what does $\int_{-1}^1 v(t) dt$ represent?

T.VI.8.006 · Common polar curves recognition pack · core · AP_CALC_BC,IB_AA_HL,IB_AA_SL

- (a) [never/neutral/1mk] $\int 6x^1 dx$.
- (b) [never/US/2mk] Evaluate $\int_{-1}^1 6x dx$.
- (c) [allowed/US/3mk ■ LAB] Desmos lab for «Common polar curves recognition pack»: vary one parameter, capture before/after, write one observation.

T.VI.8.007 · Convert polar $\leftrightarrow$ cartesian exam · core · AP_CALC_BC,IB_AA_HL,IB_AA_SL

- (a) [never/neutral/1mk] $\int 2x^1 dx$.
- (b) [never/US/2mk] Evaluate $\int_{-1}^1 2x dx$.
- (c) [allowed/US/2mk] If $v(t)$ is velocity, what does $\int_{-1}^1 v(t) dt$ represent?

T.VI.8.008 · Area between polar curves FRQ · challenge · AP_CALC_BC,IB_AA_HL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «Area between polar curves FRQ»?
- (b) [allowed/US/2mk] Write a justification sentence frame for «Area between polar curves FRQ»: Because ___, by ___, therefore ___.
- (c) [allowed/US/3mk] Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Area between polar curves FRQ», then interpret.

T.VI.8.009 · Desmos polar graphing exploration · core · AP_CALC_BC,IB_AA_HL,IB_AA_SL

- (a) [never/neutral/1mk] $\int 5x^1 dx$.
- (b) [never/US/2mk] Evaluate $\int_{-1}^1 5x dx$.
- (c) [allowed/US/3mk ■ LAB] Desmos lab for «Desmos polar graphing exploration»: vary one parameter, capture before/after, write one observation.

T.VI.8.010 · Vector-valued position/velocity/acceleration · core · AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/1mk] $a=(1, 4)$, $b=(4, 5)$. Find $a+b$ as a column vector.
- (b) [never/US/2mk] $|a|$ for $a=(1, 4)$?
- (c) [never/US/2mk] Position $A=(1,4)$, $B=(5,9)$. Vector AB ?

T.VI.8.011 · Polar to area of roses petals count · challenge · AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/1mk] $\int 5x^2 dx$.
- (b) [never/US/2mk] Evaluate $\int_{-1}^1 5x dx$.
- (c) [allowed/US/3mk ■ LAB] Desmos lab for «Polar to area of roses petals count»: vary one parameter, capture before/after, write one observation.

T.VI.8.012 · Eliminate parameter then differentiate check · core · AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/1mk] $\int 2x^1 dx$.
- (b) [never/US/2mk] Evaluate $\int_{-1}^1 2x dx$.
- (c) [allowed/US/2mk] If $v(t)$ is velocity, what does $\int_{-1}^1 v(t) dt$ represent?

T.VI.8.013 · Polar area shared regions careful bounds · challenge · AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/2mk] 150 cm to nearest 10 cm. Write the error interval for length L .
- (b) [never/US/2mk] $w=5.8$ cm, $l=10.5$ cm each to nearest 0.1 cm. Upper bound of perimeter?
- (c) [allowed/US/3mk ■ LAB] Desmos lab for «Polar area shared regions careful bounds»: vary one parameter, capture before/after, write one observation.

T.VI.8.014 · Polar/parametric FRQ segment 1 · challenge · AP_CALC_BC,IB_AA_HL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «Polar/parametric FRQ segment 1»?
- (b) [allowed/US/2mk] Write a justification sentence frame for «Polar/parametric FRQ segment 1»: Because ___, by ___, therefore ___.
- (c) [allowed/US/3mk ■ LAB] Desmos lab for «Polar/parametric FRQ segment 1»: vary one parameter, capture before/after, write one observation.

T.VI.8.015 · Polar/parametric FRQ segment 2 · challenge · AP_CALC_BC,IB_AA_HL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «Polar/parametric FRQ segment 2»?
- (b) [allowed/US/2mk] Write a justification sentence frame for «Polar/parametric FRQ segment 2»: Because ___, by ___, therefore ___.
- (c) [allowed/US/3mk ■ LAB] Desmos lab for «Polar/parametric FRQ segment 2»: vary one parameter, capture before/after, write one observation.

T.VI.8.016 · Polar/parametric FRQ segment 3 · challenge · AP_CALC_BC,IB_AA_HL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «Polar/parametric FRQ segment 3»?
- (b) [allowed/US/2mk] Write a justification sentence frame for «Polar/parametric FRQ segment 3»: Because ___, by ___, therefore ___.
- (c) [allowed/US/3mk ■ LAB] Desmos lab for «Polar/parametric FRQ segment 3»: vary one parameter, capture before/after, write one observation.

T.VI.8.017 · Polar/parametric FRQ segment 4 · challenge · AP_CALC_BC,IB_AA_HL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «Polar/parametric FRQ segment 4»?
- (b) [allowed/US/2mk] Write a justification sentence frame for «Polar/parametric FRQ segment 4»: Because ___, by ___, therefore ___.
- (c) [allowed/US/3mk ■ LAB] Desmos lab for «Polar/parametric FRQ segment 4»: vary one parameter, capture before/after, write one observation.

T.VI.9.001 · Series tests battery: nth-term first · core · AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/2mk] nth term of 2, 4, 6, 8, ...
- (b) [never/US/1mk] 20th term: $a=2$, $d=2$.
- (c) [never/US/2mk] S_n for $a=2$, $d=2$.

T.VI.9.002 · Integral test with remainder estimate · challenge · AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/2mk] 120 cm to nearest 10 cm. Write the error interval for length L.
- (b) [never/US/2mk] $w=6.2$ cm, $l=9.8$ cm each to nearest 0.1 cm. Upper bound of perimeter?
- (c) [never/US/2mk] Maximise speed = distance/time given rounded measurements. Which bounds?

T.VI.9.003 · Comparison and limit comparison drills · core · AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/2mk] Evaluate $\lim_{x \rightarrow 6} \frac{(x^2-36)}{(x-6)}$.
- (b) [allowed/US/3mk] IVT: f continuous on $[6,9]$, $f(6)=-2$, $f(9)=5$. Justify a root of $f(x)=0$.
- (c) [allowed/US/2mk] Distinguish a vertical asymptote from a removable hole in one sentence.

T.VI.9.004 · Alternating series test + error bound · core · AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/2mk] 170 cm to nearest 10 cm. Write the error interval for length L.
- (b) [never/US/2mk] $w=4.5$ cm, $l=9.3$ cm each to nearest 0.1 cm. Upper bound of perimeter?
- (c) [never/US/2mk] Maximise speed = distance/time given rounded measurements. Which bounds?

T.VI.9.005 · Ratio test radius of convergence workflow · core · AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/2mk] nth term of 8, 15, 22, 29, ...
- (b) [never/US/1mk] 20th term: $a=8$, $d=7$.
- (c) [never/US/2mk] S_n for $a=8$, $d=7$.

T.VI.9.006 · Absolute vs conditional convergence FRQ · core · AP_CALC_BC,IB_AA_HL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «Absolute vs conditional convergence FRQ»?
- (b) [allowed/US/2mk] Write a justification sentence frame for «Absolute vs conditional convergence FRQ»: Because ___, by ___, therefore ___.
- (c) [allowed/US/3mk] Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Absolute vs conditional convergence FRQ», then interpret.

T.VI.9.007 · Which test? decision tree mega-drill · challenge · AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/2mk] nth term of 7, 11, 15, 19, ...
- (b) [never/US/1mk] 20th term: $a=7$, $d=4$.
- (c) [never/US/2mk] S_n for $a=7$, $d=4$.

T.VI.9.008 · p-series and geometric recognition speed · core · AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/2mk] nth term of 2, 6, 10, 14, ...
- (b) [never/US/1mk] 20th term: $a=2$, $d=4$.
- (c) [never/US/2mk] S_n for $a=2$, $d=4$.

T.VI.9.009 · Root test when ratio inconclusive · challenge · AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/2mk] nth term of 8, 12, 16, 20, ...
- (b) [never/US/1mk] 20th term: $a=8$, $d=4$.
- (c) [never/US/2mk] S_n for $a=8$, $d=4$.

T.VI.9.010 · Telescoping series recognition · challenge · AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/2mk] nth term of 3, 7, 11, 15, ...
- (b) [never/US/1mk] 20th term: $a=3$, $d=4$.
- (c) [never/US/2mk] S_n for $a=3$, $d=4$.

T.VI.9.011 · Limit comparison with asymptotic equivalence · challenge · AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/2mk] Evaluate $\lim_{x \rightarrow 5} (x^2 - 25)/(x - 5)$.
- (b) [allowed/US/3mk] IVT: f continuous on $[5, 8]$, $f(5) = -2$, $f(8) = 5$. Justify a root of $f(x) = 0$.
- (c) [allowed/US/2mk] Distinguish a vertical asymptote from a removable hole in one sentence.

T.VI.9.012 · Series tests mixed battery 1 · challenge · AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/2mk] nth term of 4, 9, 14, 19, ...
- (b) [never/US/1mk] 20th term: $a=4$, $d=5$.
- (c) [never/US/2mk] S_n for $a=4$, $d=5$.

T.VI.9.013 · Series tests mixed battery 2 · challenge · AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/2mk] nth term of 2, 5, 8, 11, ...
- (b) [never/US/1mk] 20th term: $a=2$, $d=3$.
- (c) [never/US/2mk] S_n for $a=2$, $d=3$.

T.VI.9.014 · Series tests mixed battery 3 · challenge · AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/2mk] nth term of 5, 11, 17, 23, ...
- (b) [never/US/1mk] 20th term: $a=5$, $d=6$.
- (c) [never/US/2mk] S_n for $a=5$, $d=6$.

T.VI.9.015 · Series tests mixed battery 4 · challenge · AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/2mk] nth term of 7, 14, 21, 28, ...
- (b) [never/US/1mk] 20th term: $a=7$, $d=7$.
- (c) [never/US/2mk] S_n for $a=7$, $d=7$.

T.VI.9.016 · Series tests mixed battery 5 · challenge · AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/2mk] nth term of 8, 13, 18, 23, ...
- (b) [never/US/1mk] 20th term: $a=8$, $d=5$.
- (c) [never/US/2mk] S_n for $a=8$, $d=5$.

T.VI.10.001 · Taylor/Maclaurin polynomial construction FRQ · core · AP_CALC_BC,IB_AA_HL,IB_AA_HL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «Taylor/Maclaurin polynomial construction FRQ»?
- (b) [allowed/US/2mk] Write a justification sentence frame for «Taylor/Maclaurin polynomial construction FRQ»: Because ____, by ____, therefore ____.
- (c) [allowed/US/3mk] Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Taylor/Maclaurin polynomial construction FRQ», then interpret.

T.VI.10.002 · Lagrange error bound FRQ · challenge · AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/2mk] 80 cm to nearest 10 cm. Write the error interval for length L .
- (b) [never/US/2mk] $w=2.6$ cm, $l=9.6$ cm each to nearest 0.1 cm. Upper bound of perimeter?
- (c) [never/US/2mk] Maximise speed = distance/time given rounded measurements. Which bounds?

T.VI.10.003 · Alternating series error for Taylor · core · AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/2mk] nth term of 3, 10, 17, 24, ...
- (b) [never/US/1mk] 20th term: $a=3$, $d=7$.
- (c) [never/US/2mk] S_n for $a=3$, $d=7$.

T.VI.10.004 · Manipulate known Maclaurin series · core · AP_CALC_BC,IB_AA_HL,IB_AA_HL

- (a) [never/neutral/2mk] nth term of 7, 10, 13, 16, ...
- (b) [never/US/1mk] 20th term: $a=7$, $d=3$.
- (c) [never/US/2mk] S_n for $a=7$, $d=3$.

T.VI.10.005 · Interval of convergence endpoint checks · core · AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/2mk] nth term of 6, 10, 14, 18, ...
- (b) [never/US/1mk] 20th term: $a=6$, $d=4$.
- (c) [never/US/2mk] S_{∞} for $a=6$, $d=4$.

T.VI.10.006 · Series for e^x , $\sin x$, $\cos x$, $1/(1-x)$ fluency · core · AP_CALC_BC,IB_AA_HL,IB_AA_HL

- (a) [never/neutral/2mk] nth term of 3, 5, 7, 9, ...
- (b) [never/US/1mk] 20th term: $a=3$, $d=2$.
- (c) [never/US/2mk] S_{∞} for $a=3$, $d=2$.

T.VI.10.007 · Approximate integrals via series FRQ · core · AP_CALC_BC,IB_AA_HL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «Approximate integrals via series FRQ»?
- (b) [allowed/US/2mk] Write a justification sentence frame for «Approximate integrals via series FRQ»: Because ____, by ____, therefore ____.
- (c) [allowed/US/3mk] Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Approximate integrals via series FRQ», then interpret.

T.VI.10.008 · IB AA HL Maclaurin exam packaging · challenge · IB_AA_HL

- (a) [never/neutral/2mk] nth term of 5, 7, 9, 11, ...
- (b) [never/IB/1mk] 20th term: $a=5$, $d=2$.
- (c) [never/IB/2mk] S_{∞} for $a=5$, $d=2$.

T.VI.10.009 · Binomial series expansion $(1+x)^k$ · challenge · AP_CALC_BC,IB_AA_HL,IB_AA_HL

- (a) [never/neutral/2mk] nth term of 2, 7, 12, 17, ...
- (b) [never/US/1mk] 20th term: $a=2$, $d=5$.
- (c) [never/US/2mk] S_{∞} for $a=2$, $d=5$.

T.VI.10.010 · Taylor series for $\ln(1+x)$ and $\arctan$ · core · AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/2mk] nth term of 4, 7, 10, 13, ...
- (b) [never/US/1mk] 20th term: $a=4$, $d=3$.
- (c) [never/US/2mk] S_{∞} for $a=4$, $d=3$.

T.VI.10.011 · Error bounds choose Lagrange vs alternating · challenge · AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/2mk] 120 cm to nearest 10 cm. Write the error interval for length L.
- (b) [never/US/2mk] $w=4.2$ cm, $l=9.1$ cm each to nearest 0.1 cm. Upper bound of perimeter?
- (c) [never/US/2mk] Maximise speed = distance/time given rounded measurements. Which bounds?

T.VI.10.012 · Replace x by x^2 etc. in Maclaurin · core · AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/2mk] nth term of 4, 6, 8, 10, ...
- (b) [never/US/1mk] 20th term: $a=4$, $d=2$.
- (c) [never/US/2mk] S_{∞} for $a=4$, $d=2$.

T.VI.10.013 · Taylor error bound practice 1 · challenge · AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/2mk] 170 cm to nearest 10 cm. Write the error interval for length L.
- (b) [never/US/2mk] $w=4.6$ cm, $l=6.8$ cm each to nearest 0.1 cm. Upper bound of perimeter?
- (c) [never/US/2mk] Maximise speed = distance/time given rounded measurements. Which bounds?

T.VI.10.014 · Taylor error bound practice 2 · challenge · AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/2mk] 40 cm to nearest 10 cm. Write the error interval for length L.
- (b) [never/US/2mk] $w=4.2$ cm, $l=6.9$ cm each to nearest 0.1 cm. Upper bound of perimeter?
- (c) [never/US/2mk] Maximise speed = distance/time given rounded measurements. Which bounds?

T.VI.10.015 · Taylor error bound practice 3 · challenge · AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/2mk] 150 cm to nearest 10 cm. Write the error interval for length L.
- (b) [never/US/2mk] $w=6.0$ cm, $l=9.3$ cm each to nearest 0.1 cm. Upper bound of perimeter?
- (c) [never/US/2mk] Maximise speed = distance/time given rounded measurements. Which bounds?

T.VI.11.001 · Study design: experiment vs observational · core · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] $X \sim N(50, 5^2)$. Approx $P(45 < X < 55)$ (empirical rule)?
- (b) [allowed/US/1mk] z-score of $x=60$ ($\mu=50, \sigma=5$)?
- (c) [gdc/US/2mk] GDC: outline finding $P(X > 55)$ for $X \sim N(50, 5^2)$.

T.VI.11.002 · Sampling methods named pack · core · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] $X \sim N(70, 10^2)$. Approx $P(60 < X < 80)$ (empirical rule)?
- (b) [allowed/US/1mk] z-score of $x=90$ ($\mu=70, \sigma=10$)?
- (c) [gdc/US/2mk] GDC: outline finding $P(X > 80)$ for $X \sim N(70, 10^2)$.

T.VI.11.003 · Bias types: undercoverage, nonresponse, response · core · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] $X \sim N(70, 10^2)$. Approx $P(60 < X < 80)$ (empirical rule)?
- (b) [allowed/US/1mk] z-score of $x=90$ ($\mu=70, \sigma=10$)?
- (c) [gdc/US/2mk] GDC: outline finding $P(X > 80)$ for $X \sim N(70, 10^2)$.

T.VI.11.004 · Simulation-based inference pedagogy · challenge · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] $X \sim N(50, 5^2)$. Approx $P(45 < X < 55)$ (empirical rule)?
- (b) [allowed/US/1mk] z-score of $x=60$ ($\mu=50, \sigma=5$)?
- (c) [gdc/US/2mk] GDC: outline finding $P(X > 55)$ for $X \sim N(50, 5^2)$.

T.VI.11.005 · One-proportion z-interval procedure · core · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] $X \sim N(50, 5^2)$. Approx $P(45 < X < 55)$ (empirical rule)?
- (b) [allowed/US/1mk] z-score of $x=60$ ($\mu=50, \sigma=5$)?
- (c) [gdc/US/2mk] GDC: outline finding $P(X > 55)$ for $X \sim N(50, 5^2)$.

T.VI.11.006 · One-proportion z-test procedure · core · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] $X \sim N(70, 10^2)$. Approx $P(60 < X < 80)$ (empirical rule)?
- (b) [allowed/US/1mk] z-score of $x=90$ ($\mu=70, \sigma=10$)?
- (c) [gdc/US/2mk] GDC: outline finding $P(X > 80)$ for $X \sim N(70, 10^2)$.

T.VI.11.007 · Two-proportion z procedures · challenge · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] $X \sim N(100, 15^2)$. Approx $P(85 < X < 115)$ (empirical rule)?
- (b) [allowed/US/1mk] z-score of $x=130$ ($\mu=100, \sigma=15$)?
- (c) [gdc/US/2mk] GDC: outline finding $P(X > 115)$ for $X \sim N(100, 15^2)$.

T.VI.11.008 · One-sample t-interval for mean · core · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] $X \sim N(100, 15^2)$. Approx $P(85 < X < 115)$ (empirical rule)?
- (b) [allowed/US/1mk] z-score of $x=130$ ($\mu=100, \sigma=15$)?
- (c) [gdc/US/2mk] GDC: outline finding $P(X > 115)$ for $X \sim N(100, 15^2)$.

T.VI.11.009 · One-sample t-test for mean · core · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] $X \sim N(70, 10^2)$. Approx $P(60 < X < 80)$ (empirical rule)?
- (b) [allowed/US/1mk] z-score of $x=90$ ($\mu=70, \sigma=10$)?
- (c) [gdc/US/2mk] GDC: outline finding $P(X > 80)$ for $X \sim N(70, 10^2)$.

T.VI.11.010 · Two-sample t procedures / paired data · challenge · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] $X \sim N(70, 10^2)$. Approx $P(60 < X < 80)$ (empirical rule)?
- (b) [allowed/US/1mk] z-score of $x=90$ ($\mu=70, \sigma=10$)?
- (c) [gdc/US/2mk] GDC: outline finding $P(X > 80)$ for $X \sim N(70, 10^2)$.

T.VI.11.011 · Chi-square GOF test · core · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] $X \sim N(100, 15^2)$. Approx $P(85 < X < 115)$ (empirical rule)?
- (b) [allowed/US/1mk] z-score of $x=130$ ($\mu=100, \sigma=15$)?
- (c) [gdc/US/2mk] GDC: outline finding $P(X > 115)$ for $X \sim N(100, 15^2)$.

T.VI.11.012 · Chi-square independence / homogeneity · challenge · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] $X \sim N(70, 10^2)$. Approx $P(60 < X < 80)$ (empirical rule)?
- (b) [allowed/US/1mk] z-score of $x=90$ ($\mu=70, \sigma=10$)?
- (c) [gdc/US/2mk] GDC: outline finding $P(X > 80)$ for $X \sim N(70, 10^2)$.

T.VI.11.013 · Slope inference / CI for regression slope · challenge · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] $X \sim N(70, 10^2)$. Approx $P(60 < X < 80)$ (empirical rule)?
- (b) [allowed/US/1mk] z-score of $x=90$ ($\mu=70$, $\sigma=10$)?
- (c) [gdc/US/2mk] GDC: outline finding $P(X > 80)$ for $X \sim N(70, 10^2)$.

T.VI.11.014 · AP Stats FRQ investigative communication · challenge · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «AP Stats FRQ investigative communication»?
- (b) [allowed/US/2mk] Write a justification sentence frame for «AP Stats FRQ investigative communication»: Because ____, by ____, therefore ____.
- (c) [allowed/US/3mk] Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «AP Stats FRQ investigative communication», then interpret.

T.VI.11.015 · Formula sheet / table fluency (z, t, χ^2) · core · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] $X \sim N(70, 10^2)$. Approx $P(60 < X < 80)$ (empirical rule)?
- (b) [allowed/US/1mk] z-score of $x=90$ ($\mu=70$, $\sigma=10$)?
- (c) [gdc/US/2mk] GDC: outline finding $P(X > 80)$ for $X \sim N(70, 10^2)$.

T.VI.11.016 · IB AI inference GDC workflows · core · IB_AI_HL,IB_AI_SL

- (a) [gdc/neutral/1mk] $X \sim N(50, 5^2)$. Approx $P(45 < X < 55)$ (empirical rule)?
- (b) [gdc/IB/1mk] z-score of $x=60$ ($\mu=50$, $\sigma=5$)?
- (c) [gdc/IB/3mk ■ LAB] GDC lab for «IB AI inference GDC workflows»: record menu path and numerical output.

T.VI.11.017 · Type I / Type II errors in context · core · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] $X \sim N(70, 10^2)$. Approx $P(60 < X < 80)$ (empirical rule)?
- (b) [allowed/US/1mk] z-score of $x=90$ ($\mu=70$, $\sigma=10$)?
- (c) [gdc/US/2mk] GDC: outline finding $P(X > 80)$ for $X \sim N(70, 10^2)$.

T.VI.11.018 · Conditions check mantra before every procedure · core · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] $X \sim N(50, 5^2)$. Approx $P(45 < X < 55)$ (empirical rule)?
- (b) [allowed/US/1mk] z-score of $x=60$ ($\mu=50$, $\sigma=5$)?
- (c) [gdc/US/2mk] GDC: outline finding $P(X > 55)$ for $X \sim N(50, 5^2)$.

T.VI.11.019 · Pooling for two-proportion z-test · challenge · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] $X \sim N(50, 5^2)$. Approx $P(45 < X < 55)$ (empirical rule)?
- (b) [allowed/US/1mk] z-score of $x=60$ ($\mu=50$, $\sigma=5$)?
- (c) [gdc/US/2mk] GDC: outline finding $P(X > 55)$ for $X \sim N(50, 5^2)$.

T.VI.11.020 · df for t and χ^2 fluency · core · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] $X \sim N(100, 15^2)$. Approx $P(85 < X < 115)$ (empirical rule)?
- (b) [allowed/US/1mk] z-score of $x=130$ ($\mu=100$, $\sigma=15$)?
- (c) [gdc/US/2mk] GDC: outline finding $P(X > 115)$ for $X \sim N(100, 15^2)$.

T.VI.11.021 · Confidence level vs interval width tradeoff · core · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] $X \sim N(100, 15^2)$. Approx $P(85 < X < 115)$ (empirical rule)?
- (b) [allowed/US/1mk] z-score of $x=130$ ($\mu=100$, $\sigma=15$)?
- (c) [gdc/US/2mk] GDC: outline finding $P(X > 115)$ for $X \sim N(100, 15^2)$.

T.VI.11.022 · Writing hypotheses H_0 H_a templates · core · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] $X \sim N(50, 5^2)$. Approx $P(45 < X < 55)$ (empirical rule)?
- (b) [allowed/US/1mk] z-score of $x=60$ ($\mu=50$, $\sigma=5$)?
- (c) [gdc/US/2mk] GDC: outline finding $P(X > 55)$ for $X \sim N(50, 5^2)$.

T.VI.11.023 · Conclusion sentence frames AP style · core · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] $X \sim N(100, 15^2)$. Approx $P(85 < X < 115)$ (empirical rule)?
- (b) [allowed/US/1mk] z-score of $x=130$ ($\mu=100$, $\sigma=15$)?
- (c) [gdc/US/2mk] GDC: outline finding $P(X > 115)$ for $X \sim N(100, 15^2)$.

T.VI.11.024 · IB AI bivariate: Pearson r interpretation · core · IB_AI_SL,IB_AI_HL

- (a) [gdc/neutral/1mk] $X \sim N(50, 5^2)$. Approx $P(45 < X < 55)$ (empirical rule)?
- (b) [gdc/IB/1mk] z-score of $x=60$ ($\mu=50$, $\sigma=5$)?
- (c) [gdc/IB/3mk ■ LAB] GDC lab for «IB AI bivariate: Pearson r interpretation»: record menu path and numerical output.

T.VI.11.025 · IB AI: Spearman rank correlation (if in guide) · strange · IB_AI_HL

- (a) [allowed/neutral/1mk] $X \sim N(50, 5^2)$. Approx $P(45 < X < 55)$ (empirical rule)?
- (b) [allowed/IB/1mk] z-score of $x=60$ ($\mu=50$, $\sigma=5$)?
- (c) [gdc/IB/2mk] GDC: outline finding $P(X > 55)$ for $X \sim N(50, 5^2)$.

T.VI.11.026 · Margin of error from CI half-width · core · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] $X \sim N(70, 10^2)$. Approx $P(60 < X < 80)$ (empirical rule)?
- (b) [allowed/US/1mk] z-score of $x=90$ ($\mu=70$, $\sigma=10$)?
- (c) [gdc/US/2mk] GDC: outline finding $P(X > 80)$ for $X \sim N(70, 10^2)$.

T.VI.11.027 · AP Stats 2026 CED Unit checklist overlay · core · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] $X \sim N(70, 10^2)$. Approx $P(60 < X < 80)$ (empirical rule)?
- (b) [allowed/US/1mk] z-score of $x=90$ ($\mu=70$, $\sigma=10$)?
- (c) [gdc/US/2mk] GDC: outline finding $P(X > 80)$ for $X \sim N(70, 10^2)$.

T.VI.11.028 · Chi-square expected counts computation · core · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] $X \sim N(100, 15^2)$. Approx $P(85 < X < 115)$ (empirical rule)?
- (b) [allowed/US/1mk] z-score of $x=130$ ($\mu=100$, $\sigma=15$)?
- (c) [gdc/US/2mk] GDC: outline finding $P(X > 115)$ for $X \sim N(100, 15^2)$.

T.VI.11.029 · Blocking in experiments · core · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] $X \sim N(50, 5^2)$. Approx $P(45 < X < 55)$ (empirical rule)?
- (b) [allowed/US/1mk] z-score of $x=60$ ($\mu=50$, $\sigma=5$)?
- (c) [gdc/US/2mk] GDC: outline finding $P(X > 55)$ for $X \sim N(50, 5^2)$.

T.VI.11.030 · Interpreting slope and intercept in context · core · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] $X \sim N(70, 10^2)$. Approx $P(60 < X < 80)$ (empirical rule)?
- (b) [allowed/US/1mk] z-score of $x=90$ ($\mu=70$, $\sigma=10$)?
- (c) [gdc/US/2mk] GDC: outline finding $P(X > 80)$ for $X \sim N(70, 10^2)$.

T.VI.11.031 · Standard error vs standard deviation language · challenge · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] $X \sim N(50, 5^2)$. Approx $P(45 < X < 55)$ (empirical rule)?
- (b) [allowed/US/1mk] z-score of $x=60$ ($\mu=50$, $\sigma=5$)?
- (c) [gdc/US/2mk] GDC: outline finding $P(X > 55)$ for $X \sim N(50, 5^2)$.

T.VI.11.032 · Full inference write-up timed drills · challenge · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] $X \sim N(50, 5^2)$. Approx $P(45 < X < 55)$ (empirical rule)?
- (b) [allowed/US/1mk] z-score of $x=60$ ($\mu=50$, $\sigma=5$)?
- (c) [gdc/US/2mk] GDC: outline finding $P(X > 55)$ for $X \sim N(50, 5^2)$.

T.VI.11.033 · Power of a test conceptually · challenge · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] $X \sim N(100, 15^2)$. Approx $P(85 < X < 115)$ (empirical rule)?
- (b) [allowed/US/1mk] z-score of $x=130$ ($\mu=100$, $\sigma=15$)?
- (c) [gdc/US/2mk] GDC: outline finding $P(X > 115)$ for $X \sim N(100, 15^2)$.

T.VI.11.034 · Interpreting P-value correctly (language drills) · core · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] $X \sim N(100, 15^2)$. Approx $P(85 < X < 115)$ (empirical rule)?
- (b) [allowed/US/1mk] z-score of $x=130$ ($\mu=100$, $\sigma=15$)?
- (c) [gdc/US/2mk] GDC: outline finding $P(X > 115)$ for $X \sim N(100, 15^2)$.

T.VI.11.035 · IB AI χ^2 GDC and critical value methods · core · IB_AI_HL

- (a) [gdc/neutral/1mk] $X \sim N(100, 15^2)$. Approx $P(85 < X < 115)$ (empirical rule)?
- (b) [gdc/IB/1mk] z-score of $x=130$ ($\mu=100$, $\sigma=15$)?
- (c) [gdc/IB/3mk ■ LAB] GDC lab for «IB AI χ^2 GDC and critical value methods»: record menu path and numerical output.

T.VI.11.036 · Matched pairs design advantages · core · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] $X \sim N(70, 10^2)$. Approx $P(60 < X < 80)$ (empirical rule)?
- (b) [allowed/US/1mk] z-score of $x=90$ ($\mu=70$, $\sigma=10$)?
- (c) [gdc/US/2mk] GDC: outline finding $P(X > 80)$ for $X \sim N(70, 10^2)$.

T.VI.11.037 · AP Stats: interpreting r^2 in context · core · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] $X \sim N(100, 15^2)$. Approx $P(85 < X < 115)$ (empirical rule)?
- (b) [allowed/US/1mk] z-score of $x=130$ ($\mu=100$, $\sigma=15$)?
- (c) [gdc/US/2mk] GDC: outline finding $P(X > 115)$ for $X \sim N(100, 15^2)$.

T.VI.11.038 · AP Stats procedure write-up 1 · challenge · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] $X \sim N(50, 5^2)$. Approx $P(45 < X < 55)$ (empirical rule)?
- (b) [allowed/US/1mk] z-score of $x=60$ ($\mu=50$, $\sigma=5$)?
- (c) [gdc/US/2mk] GDC: outline finding $P(X > 55)$ for $X \sim N(50, 5^2)$.

T.VI.11.039 · AP Stats procedure write-up 2 · challenge · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] $X \sim N(50, 5^2)$. Approx $P(45 < X < 55)$ (empirical rule)?
- (b) [allowed/US/1mk] z-score of $x=60$ ($\mu=50$, $\sigma=5$)?
- (c) [gdc/US/2mk] GDC: outline finding $P(X > 55)$ for $X \sim N(50, 5^2)$.

T.VI.11.040 · AP Stats procedure write-up 3 · challenge · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] $X \sim N(50, 5^2)$. Approx $P(45 < X < 55)$ (empirical rule)?
- (b) [allowed/US/1mk] z-score of $x=60$ ($\mu=50$, $\sigma=5$)?
- (c) [gdc/US/2mk] GDC: outline finding $P(X > 55)$ for $X \sim N(50, 5^2)$.

T.VI.11.041 · AP Stats procedure write-up 4 · challenge · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] $X \sim N(50, 5^2)$. Approx $P(45 < X < 55)$ (empirical rule)?
- (b) [allowed/US/1mk] z-score of $x=60$ ($\mu=50$, $\sigma=5$)?
- (c) [gdc/US/2mk] GDC: outline finding $P(X > 55)$ for $X \sim N(50, 5^2)$.

T.VI.11.042 · AP Stats procedure write-up 5 · challenge · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] $X \sim N(70, 10^2)$. Approx $P(60 < X < 80)$ (empirical rule)?
- (b) [allowed/US/1mk] z-score of $x=90$ ($\mu=70$, $\sigma=10$)?
- (c) [gdc/US/2mk] GDC: outline finding $P(X > 80)$ for $X \sim N(70, 10^2)$.

T.VI.11.043 · IB AI inference GDC task 1 · challenge · IB_AI_HL,IB_AI_SL

- (a) [gdc/neutral/1mk] $X \sim N(50, 5^2)$. Approx $P(45 < X < 55)$ (empirical rule)?
- (b) [gdc/IB/1mk] z-score of $x=60$ ($\mu=50$, $\sigma=5$)?
- (c) [gdc/IB/3mk ■ LAB] GDC lab for «IB AI inference GDC task 1»: record menu path and numerical output.

T.VI.11.044 · IB AI inference GDC task 2 · challenge · IB_AI_HL,IB_AI_SL

- (a) [gdc/neutral/1mk] $X \sim N(100, 15^2)$. Approx $P(85 < X < 115)$ (empirical rule)?
- (b) [gdc/IB/1mk] z-score of $x=130$ ($\mu=100$, $\sigma=15$)?
- (c) [gdc/IB/3mk ■ LAB] GDC lab for «IB AI inference GDC task 2»: record menu path and numerical output.

T.VI.11.045 · IB AI inference GDC task 3 · challenge · IB_AI_HL,IB_AI_SL

- (a) [gdc/neutral/1mk] $X \sim N(100, 15^2)$. Approx $P(85 < X < 115)$ (empirical rule)?
- (b) [gdc/IB/1mk] z-score of $x=130$ ($\mu=100$, $\sigma=15$)?
- (c) [gdc/IB/3mk ■ LAB] GDC lab for «IB AI inference GDC task 3»: record menu path and numerical output.

T.VI.12.001 · SUVAT links to calculus motion FRQ · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «SUVAT links to calculus motion FRQ»?
- (b) [allowed/US/2mk] Write a justification sentence frame for «SUVAT links to calculus motion FRQ»: Because ___, by ___, therefore ___.
- (c) [allowed/US/3mk] Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «SUVAT links to calculus motion FRQ», then interpret.

T.VI.12.002 · Variable acceleration from $a(t)$ · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «Variable acceleration from $a(t)$ »?
- (b) [allowed/US/2mk] Write a justification sentence frame for «Variable acceleration from $a(t)$ »: Because ___, by ___, therefore ___.
- (c) [allowed/US/3mk] Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Variable acceleration from $a(t)$ », then interpret.

T.VI.3.015 · AP FRQ justification practice 1 · challenge · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «AP FRQ justification practice 1»?
- (b) [allowed/US/2mk] Write a justification sentence frame for «AP FRQ justification practice 1»: Because ___, by ___, therefore ___.
- (c) [allowed/US/3mk] Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «AP FRQ justification practice 1», then interpret.

T.VI.3.016 · AP FRQ justification practice 2 · challenge · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «AP FRQ justification practice 2»?
- (b) [allowed/US/2mk] Write a justification sentence frame for «AP FRQ justification practice 2»: Because ___, by ___, therefore ___.
- (c) [allowed/US/3mk] Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «AP FRQ justification practice 2», then interpret.

T.VI.3.017 · AP FRQ justification practice 3 · challenge · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «AP FRQ justification practice 3»?
- (b) [allowed/US/2mk] Write a justification sentence frame for «AP FRQ justification practice 3»: Because ___, by ___, therefore ___.
- (c) [allowed/US/3mk] Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «AP FRQ justification practice 3», then interpret.

T.VI.3.018 · AP FRQ justification practice 4 · challenge · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «AP FRQ justification practice 4»?
- (b) [allowed/US/2mk] Write a justification sentence frame for «AP FRQ justification practice 4»: Because ___, by ___, therefore ___.
- (c) [allowed/US/3mk] Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «AP FRQ justification practice 4», then interpret.

T.VI.3.019 · AP FRQ justification practice 5 · challenge · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «AP FRQ justification practice 5»?
- (b) [allowed/US/2mk] Write a justification sentence frame for «AP FRQ justification practice 5»: Because ___, by ___, therefore ___.
- (c) [allowed/US/3mk] Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «AP FRQ justification practice 5», then interpret.

T.VI.3.020 · AP FRQ justification practice 6 · challenge · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «AP FRQ justification practice 6»?
- (b) [allowed/US/2mk] Write a justification sentence frame for «AP FRQ justification practice 6»: Because ___, by ___, therefore ___.
- (c) [allowed/US/3mk] Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «AP FRQ justification practice 6», then interpret.

T.VI.3.021 · AP FRQ justification practice 7 · challenge · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «AP FRQ justification practice 7»?
- (b) [allowed/US/2mk] Write a justification sentence frame for «AP FRQ justification practice 7»: Because ___, by ___, therefore ___.
- (c) [allowed/US/3mk] Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «AP FRQ justification practice 7», then interpret.

T.VI.4.010 · FTC accumulation FRQ variant 1 · challenge · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «FTC accumulation FRQ variant 1»?
- (b) [allowed/US/2mk] Write a justification sentence frame for «FTC accumulation FRQ variant 1»: Because ___, by ___, therefore ___.
- (c) [allowed/US/3mk] ■ LAB] Graph lab for «FTC accumulation FRQ variant 1»: plot, mark key points, state the trend.

T.VI.1.011 · Limit & continuity FRQ variant 1 · challenge · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «Limit & continuity FRQ variant 1»?
- (b) [allowed/US/2mk] Write a justification sentence frame for «Limit & continuity FRQ variant 1»: Because ___, by ___, therefore ___.
- (c) [allowed/US/3mk] Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Limit & continuity FRQ variant 1», then interpret.

T.VI.4.011 · FTC accumulation FRQ variant 2 · challenge · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «FTC accumulation FRQ variant 2»?
- (b) [allowed/US/2mk] Write a justification sentence frame for «FTC accumulation FRQ variant 2»: Because ___, by ___, therefore ___.
- (c) [allowed/US/3mk ■ LAB] Graph lab for «FTC accumulation FRQ variant 2»: plot, mark key points, state the trend.

T.VI.1.012 · Limit & continuity FRQ variant 2 · challenge · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «Limit & continuity FRQ variant 2»?
- (b) [allowed/US/2mk] Write a justification sentence frame for «Limit & continuity FRQ variant 2»: Because ___, by ___, therefore ___.
- (c) [allowed/US/3mk] Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Limit & continuity FRQ variant 2», then interpret.

T.VI.4.012 · FTC accumulation FRQ variant 3 · challenge · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «FTC accumulation FRQ variant 3»?
- (b) [allowed/US/2mk] Write a justification sentence frame for «FTC accumulation FRQ variant 3»: Because ___, by ___, therefore ___.
- (c) [allowed/US/3mk ■ LAB] Graph lab for «FTC accumulation FRQ variant 3»: plot, mark key points, state the trend.

T.VI.1.013 · Limit & continuity FRQ variant 3 · challenge · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «Limit & continuity FRQ variant 3»?
- (b) [allowed/US/2mk] Write a justification sentence frame for «Limit & continuity FRQ variant 3»: Because ___, by ___, therefore ___.
- (c) [allowed/US/3mk] Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Limit & continuity FRQ variant 3», then interpret.

T.VI.2.012 · Product and quotient rule mixed AP set · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/1mk] Differentiate $y=2x^5$.
- (b) [never/US/2mk] $f(x)=2x^5$. Find $f'(2)$.
- (c) [never/US/2mk] $s(t)=2t^2$ metres. Velocity at $t=5$ s?

T.VI.3.022 · Related rates with similar triangles · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/1mk] Differentiate $y=5x^3$.
- (b) [never/US/2mk] $f(x)=5x^3$. Find $f'(2)$.
- (c) [never/US/2mk] $s(t)=5t^2$ metres. Velocity at $t=3$ s?

T.VI.3.023 · Curve sketching from calculus summary table · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/1mk] Differentiate $y=5x^5$.
- (b) [never/US/2mk] $f(x)=5x^5$. Find $f'(2)$.
- (c) [allowed/US/3mk ■ LAB] Desmos lab for «Curve sketching from calculus summary table»: vary one parameter, capture before/after, write one observation.

T.VI.4.013 · Midpoint Riemann sum vs trapezoid compare · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/1mk] $\int 3x^1 dx$.
- (b) [never/US/2mk] Evaluate $\int_1^4 3x dx$.
- (c) [allowed/US/2mk] If $v(t)$ is velocity, what does $\int_1^3 v(t) dt$ represent?

T.VI.6.010 · Washer method horizontal axis vs vertical · challenge · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/1mk] Differentiate $y=2x^5$.
- (b) [never/US/2mk] $f(x)=2x^5$. Find $f'(2)$.
- (c) [never/US/2mk] $s(t)=2t^2$ metres. Velocity at $t=5$ s?

T.VI.7.020 · Verify a proposed solution to a DE · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/1mk] Differentiate $y=5x^3$.
- (b) [never/US/2mk] $f(x)=5x^3$. Find $f'(2)$.
- (c) [never/US/2mk] $s(t)=5t^2$ metres. Velocity at $t=3$ s?

T.VI.8.018 · Speed vs velocity parametric distinction · core · AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/1mk] $\int 3x^1 dx$.
- (b) [never/US/2mk] Evaluate $\int_1^4 3x dx$.
- (c) [allowed/US/2mk] If $v(t)$ is velocity, what does $\int_1^3 v(t) dt$ represent?

T.VI.9.017 · Geometric series of functions radius · core · AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/2mk] nth term of 3, 10, 17, 24, ...
- (b) [never/US/1mk] 20th term: $a=3$, $d=7$.
- (c) [never/US/2mk] S_n for $a=3$, $d=7$.

T.VI.10.016 · Center of Taylor series not zero · challenge · AP_CALC_BC,IB_AA_HL,IB_AA_HL

- (a) [never/neutral/2mk] nth term of 8, 10, 12, 14, ...
- (b) [never/US/1mk] 20th term: $a=8$, $d=2$.
- (c) [never/US/2mk] S_n for $a=8$, $d=2$.

T.VI.11.046 · Interpreting confidence interval correctly · core · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] $X \sim N(100, 15^2)$. Approx $P(85 < X < 115)$ (empirical rule)?
- (b) [allowed/US/1mk] z-score of $x=130$ ($\mu=100$, $\sigma=15$)?
- (c) [gdc/US/2mk] GDC: outline finding $P(X > 115)$ for $X \sim N(100, 15^2)$.

T.VI.11.047 · Randomization in experiments vs sampling · core · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] $X \sim N(70, 10^2)$. Approx $P(60 < X < 80)$ (empirical rule)?
- (b) [allowed/US/1mk] z-score of $x=90$ ($\mu=70$, $\sigma=10$)?
- (c) [gdc/US/2mk] GDC: outline finding $P(X > 80)$ for $X \sim N(70, 10^2)$.

T.VI.1.014 · Limit of piecewise defined at junction · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/2mk] Evaluate $\lim_{x \rightarrow 5} (x^2 - 25)/(x - 5)$.
- (b) [allowed/US/3mk] IVT: f continuous on $[5, 8]$, $f(5) = -2$, $f(8) = 5$. Justify a root of $f(x) = 0$.
- (c) [allowed/US/2mk] Distinguish a vertical asymptote from a removable hole in one sentence.

T.VI.4.014 · Fundamental theorem verbal interpretations · core · AP_CALC_AB,AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/1mk] AP/IB style: Which theorem or definition is central to «Fundamental theorem verbal interpretations»?
- (b) [never/US/2mk] Write a justification sentence frame for «Fundamental theorem verbal interpretations»: Because ____, by ____, therefore ____.
- (c) [never/US/3mk] Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Fundamental theorem verbal interpretations», then interpret.

T.VI.8.019 · Polar intercepts and symmetry tests · core · AP_CALC_BC,IB_AA_HL

- (a) [never/neutral/1mk] $\int 2x^1 dx$.
- (b) [never/US/2mk] Evaluate $\int_{-1}^1 2x dx$.
- (c) [allowed/US/3mk] LAB Desmos lab for «Polar intercepts and symmetry tests»: vary one parameter, capture before/after, write one observation.

T.VI.10.017 · Series solution approximation of DE light · strange · AP_CALC_BC,IB_AA_HL,IB_AA_HL

- (a) [never/neutral/2mk] nth term of 2, 9, 16, 23, ...
- (b) [never/US/1mk] 20th term: $a=2$, $d=7$.
- (c) [never/US/2mk] S_n for $a=2$, $d=7$.

T.VI.11.048 · Simulation of sampling distribution of phat · challenge · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] $X \sim N(50, 5^2)$. Approx $P(45 < X < 55)$ (empirical rule)?
- (b) [allowed/US/1mk] z-score of $x=60$ ($\mu=50$, $\sigma=5$)?
- (c) [gdc/US/2mk] GDC: outline finding $P(X > 55)$ for $X \sim N(50, 5^2)$.

Answer Key — Era VI

Brief solutions. Full reasoning in machine JSON.

T.VI.1.001

- (a) The governing definition/theorem for AP FRQ: limit justification from table/graph — *Name it precisely.*
- (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*
- (c) Accept correct on-skill calculation + interpretation — *Show working + meaning.*

T.VI.1.002

- (a) 12 — *Factor and cancel.*
- (b) Sign change + continuity $\Rightarrow$ IVT guarantees c with $f(c)=0$ — *Cite IVT carefully.*
- (c) VA: $|f| \rightarrow \infty$; hole: limit exists but f undefined/ redefined — *Limit behaviour.*

T.VI.1.003

- (a) The governing definition/theorem for IVT justification FRQ stems — *Name it precisely.*
- (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*
- (c) Accept correct on-skill calculation + interpretation — *Show working + meaning.*

T.VI.1.004

- (a) The governing definition/theorem for Continuity checklist FRQ — *Name it precisely.*
- (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*
- (c) Accept correct on-skill calculation + interpretation — *Show working + meaning.*

T.VI.1.005

- (a) 6 — *Factor and cancel.*
- (b) Sign change + continuity $\Rightarrow$ IVT guarantees c with $f(c)=0$ — *Cite IVT carefully.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.VI.1.006

- (a) The governing definition/theorem for Squeeze theorem justification write-ups — *Name it precisely.*
- (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*
- (c) Accept correct on-skill calculation + interpretation — *Show working + meaning.*

T.VI.1.007

- (a) The governing definition/theorem for Horizontal asymptotes via limits at infinity FRQ — *Name it precisely.*
- (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.VI.1.008

- (a) The governing definition/theorem for Vertical asymptotes vs holes FRQ language — *Name it precisely.*
- (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*
- (c) Accept correct on-skill calculation + interpretation — *Show working + meaning.*

T.VI.1.009

- (a) 8 — *Factor and cancel.*
- (b) Sign change + continuity $\Rightarrow$ IVT guarantees c with $f(c)=0$ — *Cite IVT carefully.*
- (c) VA: $|f| \rightarrow \infty$; hole: limit exists but f undefined/ redefined — *Limit behaviour.*

T.VI.1.010

- (a) 12 — *Factor and cancel.*
- (b) Sign change + continuity $\Rightarrow$ IVT guarantees c with $f(c)=0$ — *Cite IVT carefully.*
- (c) VA: $|f| \rightarrow \infty$; hole: limit exists but f undefined/ redefined — *Limit behaviour.*

T.VI.2.001

- (a) The governing definition/theorem for AP FRQ: definition of derivative as limit — *Name it precisely.*
- (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*
- (c) Accept correct on-skill calculation + interpretation — *Show working + meaning.*

T.VI.2.002

- (a) The governing definition/theorem for Differentiability vs continuity FRQ traps — *Name it precisely.*
- (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*
- (c) Accept correct on-skill calculation + interpretation — *Show working + meaning.*

T.VI.2.003

- (a) $16x^3$ — *Power rule.*
- (b) 128 — *Evaluate derivative.*
- (c) 32 m/s — $v=ds/dt$.

T.VI.2.004

- (a) The governing definition/theorem for Implicit differentiation AP FRQ — *Name it precisely.*

- (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*
- (c) Accept correct on-skill calculation + interpretation — *Show working + meaning.*

T.VI.2.005

- (a) $10x^1$ — *Power rule.*
- (b) 20 — *Evaluate derivative.*
- (c) 20 m/s — $v=ds/dt$.

T.VI.2.006

- (a) $12x^1$ — *Power rule.*
- (b) 24 — *Evaluate derivative.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.VI.2.007

- (a) The governing definition/theorem for Chain rule layered AP items — *Name it precisely.*
- (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*
- (c) Accept correct on-skill calculation + interpretation — *Show working + meaning.*

T.VI.2.008

- (a) $28x^3$ — *Power rule.*
- (b) 224 — *Evaluate derivative.*
- (c) 56 m/s — $v=ds/dt$.

T.VI.2.009

- (a) $21x^2$ — *Power rule.*
- (b) 84 — *Evaluate derivative.*
- (c) 42 m/s — $v=ds/dt$.

T.VI.2.010

- (a) $14x^1$ — *Power rule.*
- (b) 28 — *Evaluate derivative.*
- (c) 28 m/s — $v=ds/dt$.

T.VI.2.011

- (a) The governing definition/theorem for Approximate derivative from table FRQ — *Name it precisely.*
- (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*
- (c) Accept correct on-skill calculation + interpretation — *Show working + meaning.*

T.VI.3.001

- (a) The governing definition/theorem for MVT justification FRQ stems — *Name it precisely.*
- (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*
- (c) Accept correct on-skill calculation + interpretation — *Show working + meaning.*

T.VI.3.002

- (a) The governing definition/theorem for EVT / Extreme Value Theorem justification — *Name it precisely.*
- (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*
- (c) Accept correct on-skill calculation + interpretation — *Show working + meaning.*

T.VI.3.003

- (a) $8x^1$ — *Power rule.*
- (b) 16 — *Evaluate derivative.*
- (c) 16 m/s — $v=ds/dt$.

T.VI.3.004

- (a) The governing definition/theorem for f, f', f'' sign chart FRQ classic — *Name it precisely.*
- (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.VI.3.005

- (a) The governing definition/theorem for Optimization AP contextual FRQ — *Name it precisely.*
- (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*
- (c) Accept correct on-skill calculation + interpretation — *Show working + meaning.*

T.VI.3.006

- (a) $20x^3$ — *Power rule.*
- (b) 160 — *Evaluate derivative.*
- (c) 40 m/s — $v=ds/dt$.

T.VI.3.007

- (a) The governing definition/theorem for Particle motion: velocity/acceleration FRQ — *Name it precisely.*
- (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*
- (c) Accept correct on-skill calculation + interpretation — *Show working + meaning.*

T.VI.3.008

- (a) $30x^4$ — *Power rule.*
- (b) 480 — *Evaluate derivative.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.VI.3.009

- (a) $4x^1$ — *Power rule.*
- (b) 8 — *Evaluate derivative.*
- (c) 8 m/s — $v=ds/dt$.

T.VI.3.010

- (a) The governing definition/theorem for Concavity and inflection FRQ write-ups — *Name it precisely.*
- (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.VI.3.011

- (a) $12x^3$ — *Power rule.*
- (b) 96 — *Evaluate derivative.*
- (c) 24 m/s — $v=ds/dt$.

T.VI.3.012

- (a) $12x^2$ — *Power rule.*
- (b) 48 — *Evaluate derivative.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.VI.3.013

- (a) $15x^2$ — *Power rule.*
- (b) 60 — *Evaluate derivative.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.VI.3.014

- (a) The governing definition/theorem for Absolute extrema on closed interval FRQ — *Name it precisely.*
- (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*
- (c) Accept correct on-skill calculation + interpretation — *Show working + meaning.*

T.VI.4.001

- (a) The governing definition/theorem for Riemann sum notation left/right/mid FRQ — *Name it precisely.*
- (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*
- (c) Accept correct on-skill calculation + interpretation — *Show working + meaning.*

T.VI.4.002

- (a) The governing definition/theorem for FTC graphical accumulation FRQ — *Name it precisely.*
- (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.VI.4.003

- (a) $(5/5)x^5+C$ — $+C$.
- (b) $5/2$ — *Antiderivative then FTC.*
- (c) Net displacement from $t=0$ to 3 — *Integral of rate.*

T.VI.4.004

- (a) The governing definition/theorem for Definite integral properties FRQ — *Name it precisely.*
- (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*
- (c) Accept correct on-skill calculation + interpretation — *Show working + meaning.*

T.VI.4.005

- (a) The governing definition/theorem for Trapezoidal rule approximation FRQ — *Name it precisely.*
- (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*
- (c) Accept correct on-skill calculation + interpretation — *Show working + meaning.*

T.VI.4.006

- (a) $(6/5)x^5+C$ — $+C$.
- (b) 3 — *Antiderivative then FTC.*
- (c) Net displacement from $t=0$ to 3 — *Integral of rate.*

T.VI.4.007

- (a) $(5/4)x^4+C$ — $+C$.
- (b) $5/2$ — *Antiderivative then FTC.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.VI.4.008

- (a) The governing definition/theorem for Accumulation from rate-in rate-out FRQ — *Name it precisely.*
- (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.VI.4.009

- (a) $(6/3)x^3 + C$ — $+C$.
- (b) 3 — *Antiderivative then FTC.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.VI.5.001

- (a) $(5/4)x^4 + C$ — $+C$.
- (b) $5/2$ — *Antiderivative then FTC.*
- (c) Net displacement from $t=0$ to 3 — *Integral of rate.*

T.VI.5.002

- (a) $(2/5)x^5 + C$ — $+C$.
- (b) 1 — *Antiderivative then FTC.*
- (c) Net displacement from $t=0$ to 3 — *Integral of rate.*

T.VI.5.003

- (a) $(4/5)x^5 + C$ — $+C$.
- (b) 2 — *Antiderivative then FTC.*
- (c) Net displacement from $t=0$ to 3 — *Integral of rate.*

T.VI.5.004

- (a) $(2/2)x^2 + C$ — $+C$.
- (b) 1 — *Antiderivative then FTC.*
- (c) Net displacement from $t=0$ to 3 — *Integral of rate.*

T.VI.5.005

- (a) $(4/5)x^5 + C$ — $+C$.
- (b) 2 — *Antiderivative then FTC.*
- (c) Net displacement from $t=0$ to 3 — *Integral of rate.*

T.VI.6.001

- (a) $145 \leq L < 155$ — $\pm$ half unit.
- (b) 31.20 cm — *Use UBs in $P=2(L+W)$.*
- (c) $UB(\text{distance}) \div LB(\text{time})$ — *Max quotient: max numerator, min denominator.*

T.VI.6.002

- (a) The governing definition/theorem for Disk/washer volume FRQ — *Name it precisely.*
- (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*
- (c) Accept correct on-skill calculation + interpretation — *Show working + meaning.*

T.VI.6.003

- (a) The governing definition/theorem for Known cross-section volume FRQ — *Name it precisely.*
- (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*
- (c) Accept correct on-skill calculation + interpretation — *Show working + meaning.*

T.VI.6.004

- (a) The governing definition/theorem for Net change theorem worded FRQ — *Name it precisely.*
- (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*
- (c) Accept correct on-skill calculation + interpretation — *Show working + meaning.*

T.VI.6.005

- (a) $25x^4$ — *Power rule.*
- (b) 400 — *Evaluate derivative.*
- (c) 50 m/s — $v=ds/dt$.

T.VI.6.006

- (a) $24x^3$ — *Power rule.*
- (b) 192 — *Evaluate derivative.*
- (c) 48 m/s — $v=ds/dt$.

T.VI.6.007

- (a) The governing definition/theorem for Work pumping liquid FRQ style — *Name it precisely.*
- (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*
- (c) Accept correct on-skill calculation + interpretation — *Show working + meaning.*

T.VI.6.008

- (a) $25x^4$ — *Power rule.*
- (b) 400 — *Evaluate derivative.*

(c) 50 m/s — $v=ds/dt$.

T.VI.6.009

(a) $8x^3$ — *Power rule.*

(b) 64 — *Evaluate derivative.*

(c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.VI.7.001

(a) The governing definition/theorem for Slope fields sketch and match FRQ — *Name it precisely.*

(b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*

(c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.VI.7.002

(a) The governing definition/theorem for Euler's method table FRQ — *Name it precisely.*

(b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*

(c) Accept correct on-skill calculation + interpretation — *Show working + meaning.*

T.VI.7.003

(a) $24x^3$ — *Power rule.*

(b) 192 — *Evaluate derivative.*

(c) 48 m/s — $v=ds/dt$.

T.VI.7.004

(a) The governing definition/theorem for Separable DE with initial condition FRQ — *Name it precisely.*

(b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*

(c) Accept correct on-skill calculation + interpretation — *Show working + meaning.*

T.VI.7.005

(a) The governing definition/theorem for Growth/decay DE contextual FRQ — *Name it precisely.*

(b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*

(c) Accept correct on-skill calculation + interpretation — *Show working + meaning.*

T.VI.7.006

(a) $21x^2$ — *Power rule.*

(b) 84 — *Evaluate derivative.*

(c) 42 m/s — $v=ds/dt$.

T.VI.7.007

(a) $21x^2$ — *Power rule.*

(b) 84 — *Evaluate derivative.*

(c) 42 m/s — $v=ds/dt$.

T.VI.7.008

(a) $20x^4$ — *Power rule.*

(b) 320 — *Evaluate derivative.*

(c) 40 m/s — $v=ds/dt$.

T.VI.7.009

(a) $28x^3$ — *Power rule.*

(b) 224 — *Evaluate derivative.*

(c) 56 m/s — $v=ds/dt$.

T.VI.7.010

(a) $8x^3$ — *Power rule.*

(b) 64 — *Evaluate derivative.*

(c) 16 m/s — $v=ds/dt$.

T.VI.7.011

(a) $25x^4$ — *Power rule.*

(b) 400 — *Evaluate derivative.*

(c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.VI.7.012

(a) $12x^3$ — *Power rule.*

(b) 96 — *Evaluate derivative.*

(c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.VI.7.013

(a) $8x^1$ — *Power rule.*

(b) 16 — *Evaluate derivative.*

(c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.VI.7.014

- (a) $10x^1$ — *Power rule.*
- (b) 20 — *Evaluate derivative.*
- (c) 20 m/s — $v=ds/dt$.

T.VI.7.015

- (a) $9x^2$ — *Power rule.*
- (b) 36 — *Evaluate derivative.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.VI.7.016

- (a) $8x^1$ — *Power rule.*
- (b) 16 — *Evaluate derivative.*
- (c) 16 m/s — $v=ds/dt$.

T.VI.7.017

- (a) $6x^2$ — *Power rule.*
- (b) 24 — *Evaluate derivative.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.VI.7.018

- (a) $21x^2$ — *Power rule.*
- (b) 84 — *Evaluate derivative.*
- (c) 42 m/s — $v=ds/dt$.

T.VI.7.019

- (a) $24x^3$ — *Power rule.*
- (b) 192 — *Evaluate derivative.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.VI.8.001

- (a) The governing definition/theorem for Parametric derivatives dy/dx FRQ — *Name it precisely.*
- (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*
- (c) Accept correct on-skill calculation + interpretation — *Show working + meaning.*

T.VI.8.002

- (a) The governing definition/theorem for Parametric arc length FRQ — *Name it precisely.*
- (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*
- (c) Accept correct on-skill calculation + interpretation — *Show working + meaning.*

T.VI.8.003

- (a) The governing definition/theorem for Particle path parametric motion FRQ — *Name it precisely.*
- (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*
- (c) Accept correct on-skill calculation + interpretation — *Show working + meaning.*

T.VI.8.004

- (a) $165 \leq L < 175$ — $\pm$ half unit.
- (b) 32.20 cm — *Use UBs in $P=2(L+W)$.*
- (c) $UB(\text{distance}) \div LB(\text{time})$ — *Max quotient: max numerator, min denominator.*

T.VI.8.005

- (a) $(2/4)x^4+C$ — $+C$.
- (b) 1 — *Antiderivative then FTC.*
- (c) Net displacement from $t=0$ to 3 — *Integral of rate.*

T.VI.8.006

- (a) $(6/2)x^2+C$ — $+C$.
- (b) 3 — *Antiderivative then FTC.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.VI.8.007

- (a) $(2/2)x^2+C$ — $+C$.
- (b) 1 — *Antiderivative then FTC.*
- (c) Net displacement from $t=0$ to 3 — *Integral of rate.*

T.VI.8.008

- (a) The governing definition/theorem for Area between polar curves FRQ — *Name it precisely.*
- (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*
- (c) Accept correct on-skill calculation + interpretation — *Show working + meaning.*

T.VI.8.009

- (a) $(5/2)x^2 + C$ — $+C$.
- (b) $5/2$ — *Antiderivative then FTC.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.VI.8.010

- (a) $(5, 9)$ — *Add components.*
- (b) $\sqrt{17}$ — $\sqrt{x^2 + y^2}$.
- (c) $(4, 5)$ — $B - A$.

T.VI.8.011

- (a) $(5/3)x^3 + C$ — $+C$.
- (b) $5/2$ — *Antiderivative then FTC.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.VI.8.012

- (a) $(2/2)x^2 + C$ — $+C$.
- (b) 1 — *Antiderivative then FTC.*
- (c) Net displacement from $t=0$ to 3 — *Integral of rate.*

T.VI.8.013

- (a) $145 \leq L < 155$ — $\pm$ half unit.
- (b) 32.80 cm — *Use UBs in $P=2(L+W)$.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.VI.8.014

- (a) The governing definition/theorem for Polar/parametric FRQ segment 1 — *Name it precisely.*
- (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.VI.8.015

- (a) The governing definition/theorem for Polar/parametric FRQ segment 2 — *Name it precisely.*
- (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.VI.8.016

- (a) The governing definition/theorem for Polar/parametric FRQ segment 3 — *Name it precisely.*
- (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.VI.8.017

- (a) The governing definition/theorem for Polar/parametric FRQ segment 4 — *Name it precisely.*
- (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.VI.9.001

- (a) $2n$ — *Difference 2.*
- (b) 40 — $a + (n-1)d$.
- (c) 110 — $n/2[2a + (n-1)d]$.

T.VI.9.002

- (a) $115 \leq L < 125$ — $\pm$ half unit.
- (b) 32.20 cm — *Use UBs in $P=2(L+W)$.*
- (c) $UB(\text{distance}) \div LB(\text{time})$ — *Max quotient: max numerator, min denominator.*

T.VI.9.003

- (a) 12 — *Factor and cancel.*
- (b) Sign change + continuity $\Rightarrow$ IVT guarantees c with $f(c)=0$ — *Cite IVT carefully.*
- (c) VA: $|f| \rightarrow \infty$; hole: limit exists but f undefined/ redefined — *Limit behaviour.*

T.VI.9.004

- (a) $165 \leq L < 175$ — $\pm$ half unit.
- (b) 27.80 cm — *Use UBs in $P=2(L+W)$.*
- (c) $UB(\text{distance}) \div LB(\text{time})$ — *Max quotient: max numerator, min denominator.*

T.VI.9.005

- (a) $7n + 1$ — *Difference 7.*
- (b) 141 — $a + (n-1)d$.
- (c) 395 — $n/2[2a + (n-1)d]$.

T.VI.9.006

- (a) The governing definition/theorem for Absolute vs conditional convergence FRQ — *Name it precisely.*
- (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*

(c) Accept correct on-skill calculation + interpretation — *Show working + meaning.*

T.VI.9.007

- (a) $4n + 3$ — *Difference 4.*
- (b) $83 - a + (n-1)d$.
- (c) $250 - n/2[2a + (n-1)d]$.

T.VI.9.008

- (a) $4n + -2$ — *Difference 4.*
- (b) $78 - a + (n-1)d$.
- (c) $200 - n/2[2a + (n-1)d]$.

T.VI.9.009

- (a) $4n + 4$ — *Difference 4.*
- (b) $84 - a + (n-1)d$.
- (c) $260 - n/2[2a + (n-1)d]$.

T.VI.9.010

- (a) $4n + -1$ — *Difference 4.*
- (b) $79 - a + (n-1)d$.
- (c) $210 - n/2[2a + (n-1)d]$.

T.VI.9.011

- (a) 10 — *Factor and cancel.*
- (b) Sign change + continuity $\Rightarrow$ IVT guarantees c with $f(c)=0$ — *Cite IVT carefully.*
- (c) $\forall A: |\mathbf{f}| \rightarrow \infty$; hole: limit exists but f undefined/ redefined — *Limit behaviour.*

T.VI.9.012

- (a) $5n + -1$ — *Difference 5.*
- (b) $99 - a + (n-1)d$.
- (c) $265 - n/2[2a + (n-1)d]$.

T.VI.9.013

- (a) $3n + -1$ — *Difference 3.*
- (b) $59 - a + (n-1)d$.
- (c) $155 - n/2[2a + (n-1)d]$.

T.VI.9.014

- (a) $6n + -1$ — *Difference 6.*
- (b) $119 - a + (n-1)d$.
- (c) $320 - n/2[2a + (n-1)d]$.

T.VI.9.015

- (a) $7n$ — *Difference 7.*
- (b) $140 - a + (n-1)d$.
- (c) $385 - n/2[2a + (n-1)d]$.

T.VI.9.016

- (a) $5n + 3$ — *Difference 5.*
- (b) $103 - a + (n-1)d$.
- (c) $305 - n/2[2a + (n-1)d]$.

T.VI.10.001

- (a) The governing definition/theorem for Taylor/Maclaurin polynomial construction FRQ — *Name it precisely.*
- (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*
- (c) Accept correct on-skill calculation + interpretation — *Show working + meaning.*

T.VI.10.002

- (a) $75 \leq L < 85$ — $\pm$ *half unit.*
- (b) 24.60 cm — *Use UBs in $P=2(L+W)$.*
- (c) $UB(\text{distance}) \div LB(\text{time})$ — *Max quotient: max numerator, min denominator.*

T.VI.10.003

- (a) $7n + -4$ — *Difference 7.*
- (b) $136 - a + (n-1)d$.
- (c) $345 - n/2[2a + (n-1)d]$.

T.VI.10.004

- (a) $3n + 4$ — *Difference 3.*
- (b) $64 - a + (n-1)d$.
- (c) $205 - n/2[2a + (n-1)d]$.

T.VI.10.005

- (a) $4n + 2$ — *Difference 4.*
- (b) $82 - a + (n-1)d$.
- (c) $240 - n/2[2a + (n-1)d]$.

T.VI.10.006

- (a) $2n + 1$ — *Difference 2.*
- (b) $41 - a + (n-1)d$.
- (c) $120 - n/2[2a + (n-1)d]$.

T.VI.10.007

- (a) The governing definition/theorem for Approximate integrals via series FRQ — *Name it precisely.*
- (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*
- (c) Accept correct on-skill calculation + interpretation — *Show working + meaning.*

T.VI.10.008

- (a) $2n + 3$ — *Difference 2.*
- (b) $43 - a + (n-1)d$.
- (c) $140 - n/2[2a + (n-1)d]$.

T.VI.10.009

- (a) $5n + -3$ — *Difference 5.*
- (b) $97 - a + (n-1)d$.
- (c) $245 - n/2[2a + (n-1)d]$.

T.VI.10.010

- (a) $3n + 1$ — *Difference 3.*
- (b) $61 - a + (n-1)d$.
- (c) $175 - n/2[2a + (n-1)d]$.

T.VI.10.011

- (a) $115 \leq L < 125$ — $\pm$ *half unit.*
- (b) 26.80 cm — *Use UBs in $P=2(L+W)$.*
- (c) $UB(\text{distance}) \div LB(\text{time})$ — *Max quotient: max numerator, min denominator.*

T.VI.10.012

- (a) $2n + 2$ — *Difference 2.*
- (b) $42 - a + (n-1)d$.
- (c) $130 - n/2[2a + (n-1)d]$.

T.VI.10.013

- (a) $165 \leq L < 175$ — $\pm$ *half unit.*
- (b) 23.00 cm — *Use UBs in $P=2(L+W)$.*
- (c) $UB(\text{distance}) \div LB(\text{time})$ — *Max quotient: max numerator, min denominator.*

T.VI.10.014

- (a) $35 \leq L < 45$ — $\pm$ *half unit.*
- (b) 22.40 cm — *Use UBs in $P=2(L+W)$.*
- (c) $UB(\text{distance}) \div LB(\text{time})$ — *Max quotient: max numerator, min denominator.*

T.VI.10.015

- (a) $145 \leq L < 155$ — $\pm$ *half unit.*
- (b) 30.80 cm — *Use UBs in $P=2(L+W)$.*
- (c) $UB(\text{distance}) \div LB(\text{time})$ — *Max quotient: max numerator, min denominator.*

T.VI.11.001

- (a) $\approx 68\%$ — *1 SD rule.*
- (b) $2 - (x - \mu)/\sigma$.
- (c) $\text{normalcdf}(\mu + \sigma, \infty, \mu, \sigma) \approx 16\%$ — *Right-tail normalcdf.*

T.VI.11.002

- (a) $\approx 68\%$ — *1 SD rule.*
- (b) $2 - (x - \mu)/\sigma$.
- (c) $\text{normalcdf}(\mu + \sigma, \infty, \mu, \sigma) \approx 16\%$ — *Right-tail normalcdf.*

T.VI.11.003

- (a) $\approx 68\%$ — *1 SD rule.*
- (b) $2 - (x - \mu)/\sigma$.
- (c) $\text{normalcdf}(\mu + \sigma, \infty, \mu, \sigma) \approx 16\%$ — *Right-tail normalcdf.*

T.VI.11.004

- (a) $\approx 68\%$ — 1 SD rule.
- (b) $2 - (x - \mu)/\sigma$.
- (c) $\text{normalcdf}(\mu + \sigma, \infty, \mu, \sigma) \approx 16\%$ — Right-tail normalcdf.

T.VI.11.005

- (a) $\approx 68\%$ — 1 SD rule.
- (b) $2 - (x - \mu)/\sigma$.
- (c) $\text{normalcdf}(\mu + \sigma, \infty, \mu, \sigma) \approx 16\%$ — Right-tail normalcdf.

T.VI.11.006

- (a) $\approx 68\%$ — 1 SD rule.
- (b) $2 - (x - \mu)/\sigma$.
- (c) $\text{normalcdf}(\mu + \sigma, \infty, \mu, \sigma) \approx 16\%$ — Right-tail normalcdf.

T.VI.11.007

- (a) $\approx 68\%$ — 1 SD rule.
- (b) $2 - (x - \mu)/\sigma$.
- (c) $\text{normalcdf}(\mu + \sigma, \infty, \mu, \sigma) \approx 16\%$ — Right-tail normalcdf.

T.VI.11.008

- (a) $\approx 68\%$ — 1 SD rule.
- (b) $2 - (x - \mu)/\sigma$.
- (c) $\text{normalcdf}(\mu + \sigma, \infty, \mu, \sigma) \approx 16\%$ — Right-tail normalcdf.

T.VI.11.009

- (a) $\approx 68\%$ — 1 SD rule.
- (b) $2 - (x - \mu)/\sigma$.
- (c) $\text{normalcdf}(\mu + \sigma, \infty, \mu, \sigma) \approx 16\%$ — Right-tail normalcdf.

T.VI.11.010

- (a) $\approx 68\%$ — 1 SD rule.
- (b) $2 - (x - \mu)/\sigma$.
- (c) $\text{normalcdf}(\mu + \sigma, \infty, \mu, \sigma) \approx 16\%$ — Right-tail normalcdf.

T.VI.11.011

- (a) $\approx 68\%$ — 1 SD rule.
- (b) $2 - (x - \mu)/\sigma$.
- (c) $\text{normalcdf}(\mu + \sigma, \infty, \mu, \sigma) \approx 16\%$ — Right-tail normalcdf.

T.VI.11.012

- (a) $\approx 68\%$ — 1 SD rule.
- (b) $2 - (x - \mu)/\sigma$.
- (c) $\text{normalcdf}(\mu + \sigma, \infty, \mu, \sigma) \approx 16\%$ — Right-tail normalcdf.

T.VI.11.013

- (a) $\approx 68\%$ — 1 SD rule.
- (b) $2 - (x - \mu)/\sigma$.
- (c) $\text{normalcdf}(\mu + \sigma, \infty, \mu, \sigma) \approx 16\%$ — Right-tail normalcdf.

T.VI.11.014

- (a) The governing definition/theorem for AP Stats FRQ investigative communication — *Name it precisely.*
- (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*
- (c) Accept correct on-skill calculation + interpretation — *Show working + meaning.*

T.VI.11.015

- (a) $\approx 68\%$ — 1 SD rule.
- (b) $2 - (x - \mu)/\sigma$.
- (c) $\text{normalcdf}(\mu + \sigma, \infty, \mu, \sigma) \approx 16\%$ — Right-tail normalcdf.

T.VI.11.016

- (a) $\approx 68\%$ — 1 SD rule.
- (b) $2 - (x - \mu)/\sigma$.
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.VI.11.017

- (a) $\approx 68\%$ — 1 SD rule.
- (b) $2 - (x - \mu)/\sigma$.
- (c) $\text{normalcdf}(\mu + \sigma, \infty, \mu, \sigma) \approx 16\%$ — Right-tail normalcdf.

T.VI.11.018

- (a) $\approx 68\%$ — 1 SD rule.
- (b) $2 - (x - \mu)/\sigma$.

(c) $\text{normalcdf}(\mu+\sigma, \infty, \mu, \sigma) \approx 16\%$ — *Right-tail normalcdf.*

T.VI.11.019

(a) $\approx 68\%$ — *1 SD rule.*

(b) $2 - (x-\mu)/\sigma$.

(c) $\text{normalcdf}(\mu+\sigma, \infty, \mu, \sigma) \approx 16\%$ — *Right-tail normalcdf.*

T.VI.11.020

(a) $\approx 68\%$ — *1 SD rule.*

(b) $2 - (x-\mu)/\sigma$.

(c) $\text{normalcdf}(\mu+\sigma, \infty, \mu, \sigma) \approx 16\%$ — *Right-tail normalcdf.*

T.VI.11.021

(a) $\approx 68\%$ — *1 SD rule.*

(b) $2 - (x-\mu)/\sigma$.

(c) $\text{normalcdf}(\mu+\sigma, \infty, \mu, \sigma) \approx 16\%$ — *Right-tail normalcdf.*

T.VI.11.022

(a) $\approx 68\%$ — *1 SD rule.*

(b) $2 - (x-\mu)/\sigma$.

(c) $\text{normalcdf}(\mu+\sigma, \infty, \mu, \sigma) \approx 16\%$ — *Right-tail normalcdf.*

T.VI.11.023

(a) $\approx 68\%$ — *1 SD rule.*

(b) $2 - (x-\mu)/\sigma$.

(c) $\text{normalcdf}(\mu+\sigma, \infty, \mu, \sigma) \approx 16\%$ — *Right-tail normalcdf.*

T.VI.11.024

(a) $\approx 68\%$ — *1 SD rule.*

(b) $2 - (x-\mu)/\sigma$.

(c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.VI.11.025

(a) $\approx 68\%$ — *1 SD rule.*

(b) $2 - (x-\mu)/\sigma$.

(c) $\text{normalcdf}(\mu+\sigma, \infty, \mu, \sigma) \approx 16\%$ — *Right-tail normalcdf.*

T.VI.11.026

(a) $\approx 68\%$ — *1 SD rule.*

(b) $2 - (x-\mu)/\sigma$.

(c) $\text{normalcdf}(\mu+\sigma, \infty, \mu, \sigma) \approx 16\%$ — *Right-tail normalcdf.*

T.VI.11.027

(a) $\approx 68\%$ — *1 SD rule.*

(b) $2 - (x-\mu)/\sigma$.

(c) $\text{normalcdf}(\mu+\sigma, \infty, \mu, \sigma) \approx 16\%$ — *Right-tail normalcdf.*

T.VI.11.028

(a) $\approx 68\%$ — *1 SD rule.*

(b) $2 - (x-\mu)/\sigma$.

(c) $\text{normalcdf}(\mu+\sigma, \infty, \mu, \sigma) \approx 16\%$ — *Right-tail normalcdf.*

T.VI.11.029

(a) $\approx 68\%$ — *1 SD rule.*

(b) $2 - (x-\mu)/\sigma$.

(c) $\text{normalcdf}(\mu+\sigma, \infty, \mu, \sigma) \approx 16\%$ — *Right-tail normalcdf.*

T.VI.11.030

(a) $\approx 68\%$ — *1 SD rule.*

(b) $2 - (x-\mu)/\sigma$.

(c) $\text{normalcdf}(\mu+\sigma, \infty, \mu, \sigma) \approx 16\%$ — *Right-tail normalcdf.*

T.VI.11.031

(a) $\approx 68\%$ — *1 SD rule.*

(b) $2 - (x-\mu)/\sigma$.

(c) $\text{normalcdf}(\mu+\sigma, \infty, \mu, \sigma) \approx 16\%$ — *Right-tail normalcdf.*

T.VI.11.032

(a) $\approx 68\%$ — *1 SD rule.*

(b) $2 - (x-\mu)/\sigma$.

(c) $\text{normalcdf}(\mu+\sigma, \infty, \mu, \sigma) \approx 16\%$ — *Right-tail normalcdf.*

T.VI.11.033

- (a) $\approx 68\%$ — 1 SD rule.
- (b) $2 - (x - \mu)/\sigma$.
- (c) $\text{normalcdf}(\mu + \sigma, \infty, \mu, \sigma) \approx 16\%$ — Right-tail normalcdf.

T.VI.11.034

- (a) $\approx 68\%$ — 1 SD rule.
- (b) $2 - (x - \mu)/\sigma$.
- (c) $\text{normalcdf}(\mu + \sigma, \infty, \mu, \sigma) \approx 16\%$ — Right-tail normalcdf.

T.VI.11.035

- (a) $\approx 68\%$ — 1 SD rule.
- (b) $2 - (x - \mu)/\sigma$.
- (c) Lab evidence (notes/screenshot) + one verified check — Complete lab steps.

T.VI.11.036

- (a) $\approx 68\%$ — 1 SD rule.
- (b) $2 - (x - \mu)/\sigma$.
- (c) $\text{normalcdf}(\mu + \sigma, \infty, \mu, \sigma) \approx 16\%$ — Right-tail normalcdf.

T.VI.11.037

- (a) $\approx 68\%$ — 1 SD rule.
- (b) $2 - (x - \mu)/\sigma$.
- (c) $\text{normalcdf}(\mu + \sigma, \infty, \mu, \sigma) \approx 16\%$ — Right-tail normalcdf.

T.VI.11.038

- (a) $\approx 68\%$ — 1 SD rule.
- (b) $2 - (x - \mu)/\sigma$.
- (c) $\text{normalcdf}(\mu + \sigma, \infty, \mu, \sigma) \approx 16\%$ — Right-tail normalcdf.

T.VI.11.039

- (a) $\approx 68\%$ — 1 SD rule.
- (b) $2 - (x - \mu)/\sigma$.
- (c) $\text{normalcdf}(\mu + \sigma, \infty, \mu, \sigma) \approx 16\%$ — Right-tail normalcdf.

T.VI.11.040

- (a) $\approx 68\%$ — 1 SD rule.
- (b) $2 - (x - \mu)/\sigma$.
- (c) $\text{normalcdf}(\mu + \sigma, \infty, \mu, \sigma) \approx 16\%$ — Right-tail normalcdf.

T.VI.11.041

- (a) $\approx 68\%$ — 1 SD rule.
- (b) $2 - (x - \mu)/\sigma$.
- (c) $\text{normalcdf}(\mu + \sigma, \infty, \mu, \sigma) \approx 16\%$ — Right-tail normalcdf.

T.VI.11.042

- (a) $\approx 68\%$ — 1 SD rule.
- (b) $2 - (x - \mu)/\sigma$.
- (c) $\text{normalcdf}(\mu + \sigma, \infty, \mu, \sigma) \approx 16\%$ — Right-tail normalcdf.

T.VI.11.043

- (a) $\approx 68\%$ — 1 SD rule.
- (b) $2 - (x - \mu)/\sigma$.
- (c) Lab evidence (notes/screenshot) + one verified check — Complete lab steps.

T.VI.11.044

- (a) $\approx 68\%$ — 1 SD rule.
- (b) $2 - (x - \mu)/\sigma$.
- (c) Lab evidence (notes/screenshot) + one verified check — Complete lab steps.

T.VI.11.045

- (a) $\approx 68\%$ — 1 SD rule.
- (b) $2 - (x - \mu)/\sigma$.
- (c) Lab evidence (notes/screenshot) + one verified check — Complete lab steps.

T.VI.12.001

- (a) The governing definition/theorem for SUVAT links to calculus motion FRQ — Name it precisely.
- (b) Because [hypothesis], by [theorem], therefore [claim]. — FRQ communication.
- (c) Accept correct on-skill calculation + interpretation — Show working + meaning.

T.VI.12.002

- (a) The governing definition/theorem for Variable acceleration from $a(t)$ — *Name it precisely.*
- (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*
- (c) Accept correct on-skill calculation + interpretation — *Show working + meaning.*

T.VI.3.015

- (a) The governing definition/theorem for AP FRQ justification practice 1 — *Name it precisely.*
- (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*
- (c) Accept correct on-skill calculation + interpretation — *Show working + meaning.*

T.VI.3.016

- (a) The governing definition/theorem for AP FRQ justification practice 2 — *Name it precisely.*
- (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*
- (c) Accept correct on-skill calculation + interpretation — *Show working + meaning.*

T.VI.3.017

- (a) The governing definition/theorem for AP FRQ justification practice 3 — *Name it precisely.*
- (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*
- (c) Accept correct on-skill calculation + interpretation — *Show working + meaning.*

T.VI.3.018

- (a) The governing definition/theorem for AP FRQ justification practice 4 — *Name it precisely.*
- (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*
- (c) Accept correct on-skill calculation + interpretation — *Show working + meaning.*

T.VI.3.019

- (a) The governing definition/theorem for AP FRQ justification practice 5 — *Name it precisely.*
- (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*
- (c) Accept correct on-skill calculation + interpretation — *Show working + meaning.*

T.VI.3.020

- (a) The governing definition/theorem for AP FRQ justification practice 6 — *Name it precisely.*
- (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*
- (c) Accept correct on-skill calculation + interpretation — *Show working + meaning.*

T.VI.3.021

- (a) The governing definition/theorem for AP FRQ justification practice 7 — *Name it precisely.*
- (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*
- (c) Accept correct on-skill calculation + interpretation — *Show working + meaning.*

T.VI.4.010

- (a) The governing definition/theorem for FTC accumulation FRQ variant 1 — *Name it precisely.*
- (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.VI.1.011

- (a) The governing definition/theorem for Limit & continuity FRQ variant 1 — *Name it precisely.*
- (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*
- (c) Accept correct on-skill calculation + interpretation — *Show working + meaning.*

T.VI.4.011

- (a) The governing definition/theorem for FTC accumulation FRQ variant 2 — *Name it precisely.*
- (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.VI.1.012

- (a) The governing definition/theorem for Limit & continuity FRQ variant 2 — *Name it precisely.*
- (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*
- (c) Accept correct on-skill calculation + interpretation — *Show working + meaning.*

T.VI.4.012

- (a) The governing definition/theorem for FTC accumulation FRQ variant 3 — *Name it precisely.*
- (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.VI.1.013

- (a) The governing definition/theorem for Limit & continuity FRQ variant 3 — *Name it precisely.*
- (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*
- (c) Accept correct on-skill calculation + interpretation — *Show working + meaning.*

T.VI.2.012

- (a) $10x^4$ — *Power rule.*
- (b) 160 — *Evaluate derivative.*

(c) 20 m/s — $v=ds/dt$.

T.VI.3.022

- (a) $15x^2$ — *Power rule.*
- (b) 60 — *Evaluate derivative.*
- (c) 30 m/s — $v=ds/dt$.

T.VI.3.023

- (a) $25x^4$ — *Power rule.*
- (b) 400 — *Evaluate derivative.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.VI.4.013

- (a) $(3/2)x^2+C$ — $+C$.
- (b) $3/2$ — *Antiderivative then FTC.*
- (c) Net displacement from $t=0$ to 3 — *Integral of rate.*

T.VI.6.010

- (a) $10x^4$ — *Power rule.*
- (b) 160 — *Evaluate derivative.*
- (c) 20 m/s — $v=ds/dt$.

T.VI.7.020

- (a) $15x^2$ — *Power rule.*
- (b) 60 — *Evaluate derivative.*
- (c) 30 m/s — $v=ds/dt$.

T.VI.8.018

- (a) $(3/2)x^2+C$ — $+C$.
- (b) $3/2$ — *Antiderivative then FTC.*
- (c) Net displacement from $t=0$ to 3 — *Integral of rate.*

T.VI.9.017

- (a) $7n + -4$ — *Difference 7.*
- (b) 136 — $a+(n-1)d$.
- (c) 345 — $n/2[2a+(n-1)d]$.

T.VI.10.016

- (a) $2n + 6$ — *Difference 2.*
- (b) 46 — $a+(n-1)d$.
- (c) 170 — $n/2[2a+(n-1)d]$.

T.VI.11.046

- (a) $\approx 68\%$ — *1 SD rule.*
- (b) $2 - (x-\mu)/\sigma$.
- (c) $\text{normalcdf}(\mu+\sigma, \infty, \mu, \sigma) \approx 16\%$ — *Right-tail normalcdf.*

T.VI.11.047

- (a) $\approx 68\%$ — *1 SD rule.*
- (b) $2 - (x-\mu)/\sigma$.
- (c) $\text{normalcdf}(\mu+\sigma, \infty, \mu, \sigma) \approx 16\%$ — *Right-tail normalcdf.*

T.VI.1.014

- (a) 10 — *Factor and cancel.*
- (b) Sign change + continuity $\Rightarrow$ IVT guarantees c with $f(c)=0$ — *Cite IVT carefully.*
- (c) VA: $|f| \rightarrow \infty$; hole: limit exists but f undefined/ redefined — *Limit behaviour.*

T.VI.4.014

- (a) The governing definition/theorem for Fundamental theorem verbal interpretations — *Name it precisely.*
- (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*
- (c) Accept correct on-skill calculation + interpretation — *Show working + meaning.*

T.VI.8.019

- (a) $(2/2)x^2+C$ — $+C$.
- (b) 1 — *Antiderivative then FTC.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.VI.10.017

- (a) $7n + -5$ — *Difference 7.*
- (b) 135 — $a+(n-1)d$.
- (c) 335 — $n/2[2a+(n-1)d]$.

T.VI.11.048

(a) $\approx 68\%$ — *1 SD rule*.

(b) $2 - (x - \mu)/\sigma$.

(c) $\text{normalcdf}(\mu + \sigma, \infty, \mu, \sigma) \approx 16\%$ — *Right-tail normalcdf*.